BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 414 (JULY 2025)

Electrical Service Design

Final Project Report

Section: B2 Group: 01

Integrated Electrical Infrastructure Design For a 10-storey Residential Building

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Academic Honesty Statement:

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"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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1. Abstract

In this project, we designed a basic electrical system using AutoCAD for a 10-storey residential building with a ground and basement for parking, which has dimensions of 80′ −43'x 6”, and of 2 units. For electrical design purposes, we have installed electrical fittings and conduits for both normal and emergency operations. We also designed switchboards for both floor level and building level, and used a 500k VA transformer, a generator, and a 174k VAR PFI plant for power factor correction. We have also added rooftop electrical protection to prevent adverse effects of thunderstorms on the building.

2. Introduction

The objective of this project is to ensure a safe, efficient, and well-organized electrical infrastructure for the entire building. This project focuses on the comprehensive electrical design of a 10-storied residential building, developed using AutoCAD to ensure precision and clarity in layout planning.

TYPE	HIGH RISE RESIDENTIAL BUILDING
LENGTH	80’-6”
WIDTH	43’
AREA	3461.5 sq. ft
NO. OF FLOORS	12 (10 Typical, 1 Ground, 1 Basement)
UNITS PER FLOOR	2 units

Table: Building specifications

It encompasses a detailed and systematic approach to designing the electrical infrastructure for the entire building, including the basement, ground floor, and all upper floors. The design covers essential aspects such as lighting arrangements, power distribution networks, efficient wiring systems, lightning protection system tailored to meet residential needs.

3. Design

3.1 Problem Formulation (PO(b))

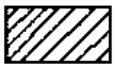
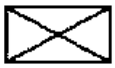
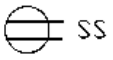
3.1.1 Literature Review

This work is based on the Bangladesh National Building Code (BNBC) 2020, which defines standards for electrical installation, safety, load calculation, and wiring. Additionally, Professor Yeasir Arafat's lecture materials provide insightful information about practical design approaches

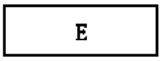
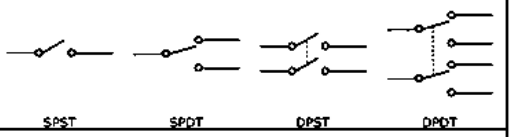
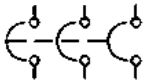
that adhere to BNBC requirements. Lighting design follows standard concepts such as illumination level, UF, and LLF, while AutoCAD resources guide schematic and layout planning. Rooftop protection uses the rolling sphere method for lightning safety. Overall, it ensures compliance with established engineering practices and standards.

3.1.2 Analysis

The project evaluates the effectiveness, safety, and efficiency of the electrical design for a ten-story residential building. It verifies lighting levels, load calculations, equipment sizing, and conduit layout to ensure safety and reliability, including emergency and lightning protection systems. Legends are used to represent all components such as fittings, conduits, switchboards, breakers, transformers, generators, and PFI plant.

Description	Height	Caption	Symbol
Wall Mounted Light	Lintel	LL	
Ceiling Light	Ceiling	CL	
Wall Mounted Tube Light	Lintel	TL	
Fan (56" diameter)	Ceiling	F	
Switch Board	Mid wall	SB	
Sub Distribution Board	Mid wall	SDB	
Main Distribution Board	Mid wall	MDB	
Exhaust Fan (8" diameter)	Lintel	EF	
2 Pin Socket	Mid wall	SS	
2 Pin TV Socket	Lower	TS	
Antenna Socket	Skirting	T	

Switchboard and others

Description	Symbol
Switch	
Energy Meter	
SPDT Two Way Switch for Automatic Transfer	
Fan Regulator	
Single Pole Circuit Breaker (SP MCCB)	
Triple Pole Circuit Breaker (TP MCCB)	
Delta to Wye Transformer	
Power Factor Improvement (PFI) Plant	
Generator	G

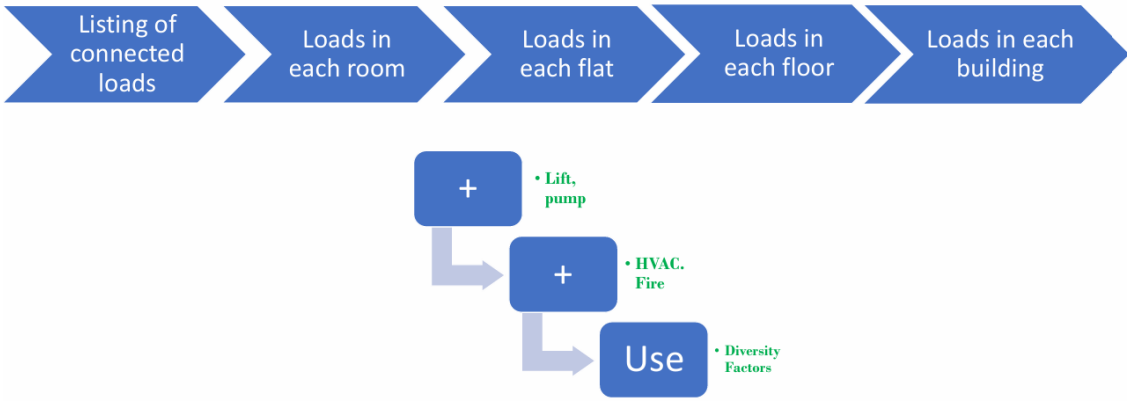
3.2 Design Method (PO(a))

The design follows a step-by-step approach to estimate and distribute loads, starting from rooms to flats, floors, and the entire building, including major loads like elevators, pumps,

HVAC, and fire systems. Diversity factors are applied to reflect realistic usage and avoid overestimation.

The same method extends to multiple buildings, considering shared loads and substations. This ensures a reliable, scalable, and cost-effective electrical design aligned with standard practices.

The flowchart for overall design and connection is given below –



3.3 AUTOCAD Design

In this project, we have designed all the layout and electrical system using AutoCAD 2026.

All over this building, we have taken a wall of 10 ″ , column of 1'×1' . We have used a variety of windows and doors depending on the type of room.

As it is a 10-storey building, alongside the stairs, we used a 4′×5 ′ lift, an open space of 4′×8 ′ -2 ″

In the typical floor, at one unit, we have added two master bedrooms of dimension 15′ − 8″×13' -8 ″ ; two bedrooms of dimension 13′ − 9″×10' -8 ″ and 10′ − 0″×13' -8 ″ ; three toilets of dimension 4′×8' - 4 ″ , 4′×7 ′ and 4′×6' -2 ″ respectively; two verandas of dimension 9′×4' and the other two verandas of dimension 13′ − 9″× 2′ − 6″ and 7′ − 4″× 3′ − 8″ ; kitchen of dimension 8′ − 0″× 13′ − 8″ , Living + Dining of dimension 39′ − 2″× 13′ − 0″ .

The other unit has almost the same dimensions as the previous unit, but in this unit, the bedroom and its embedded veranda have the dimensions of 8′ − 3″× 10′ − 8″ and 8′ − 3″× 2′ − 6″ respectively.

By taking all the measurements and placing them , we get the following design .

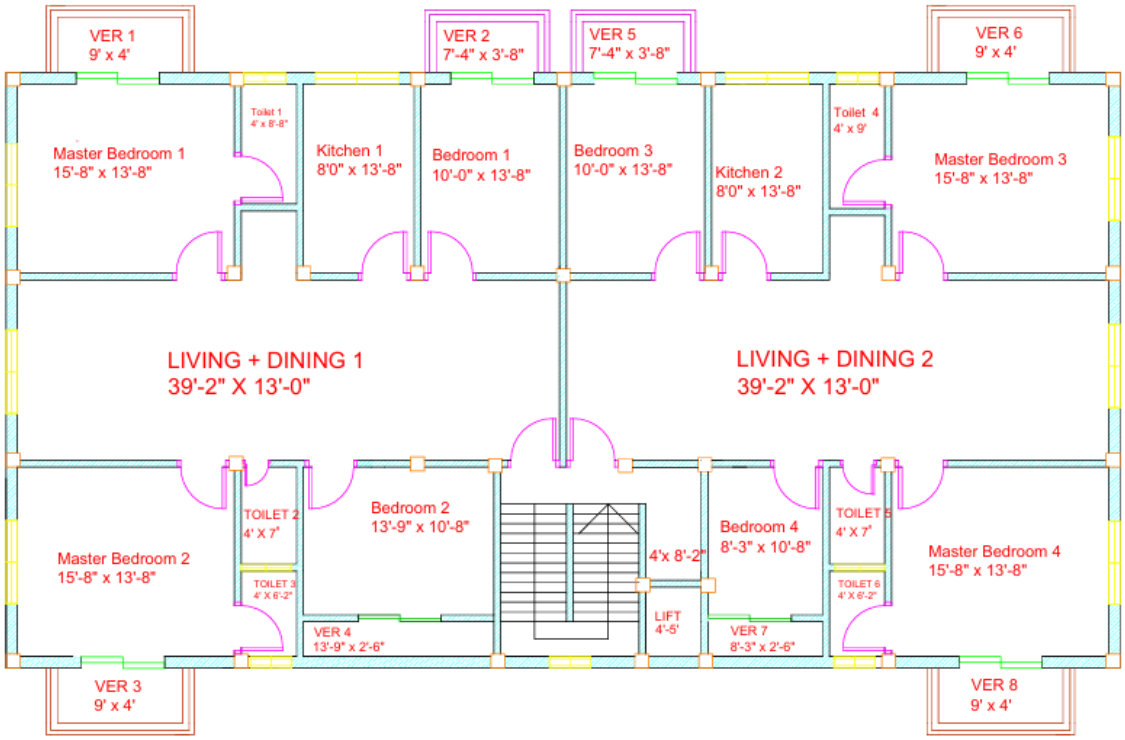


Figure: Main floor layout

On the Ground floor, we have added a Generator Room, transformer Room, Electric Switchboard Room, Guard Room, kitchen, and Toilet.

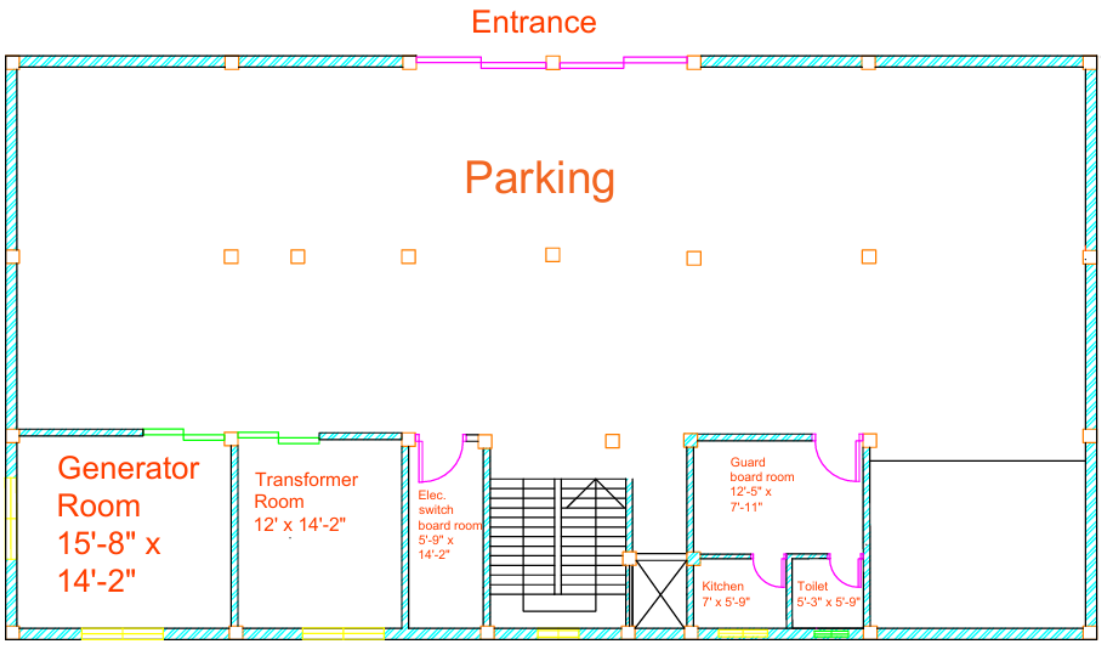


Figure: Ground floor layout

We have used the basement for parking just like the design below.

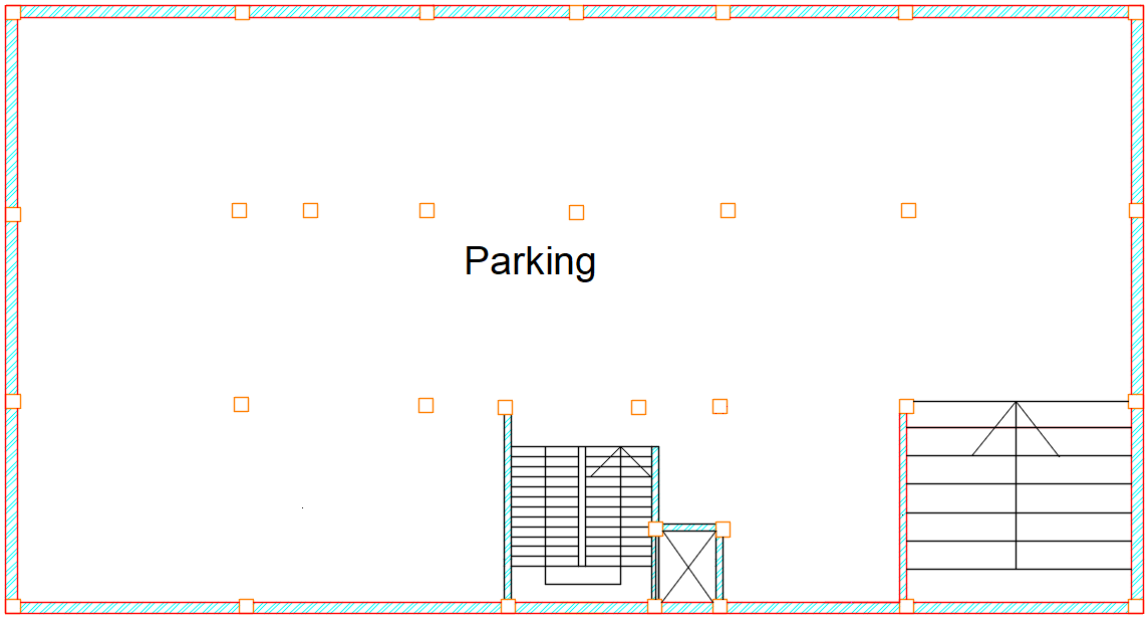


Figure: Basement layout

And, a rooftop just as the design below.

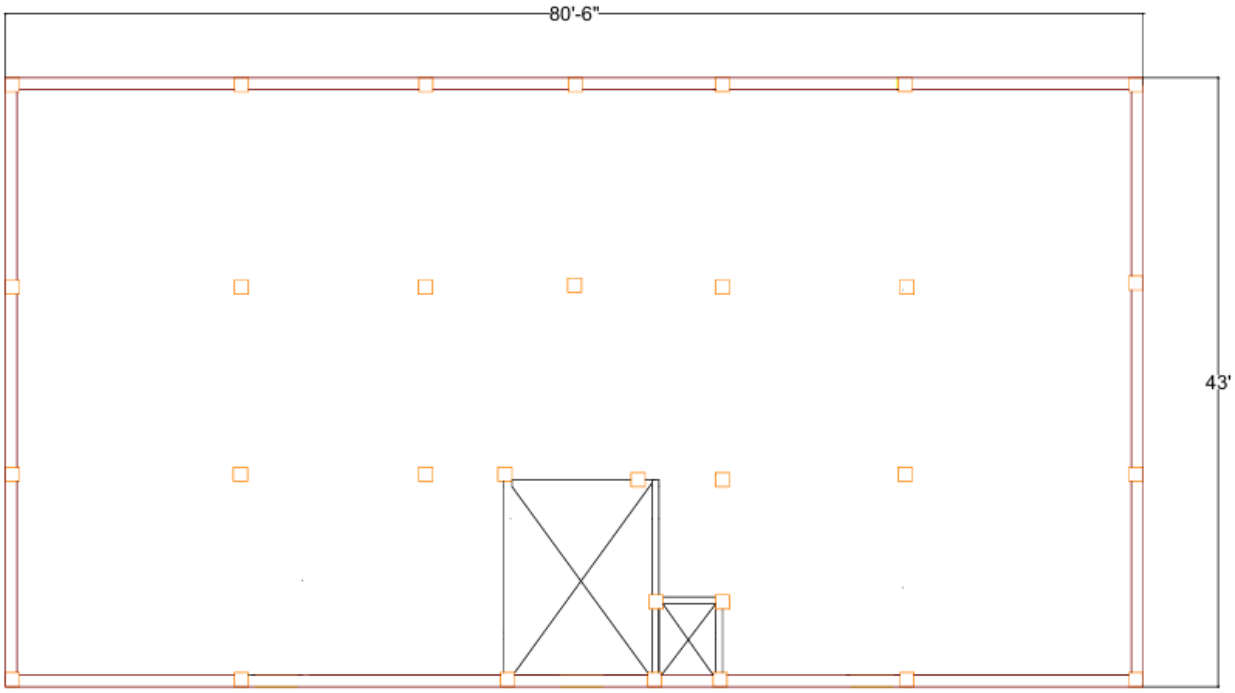


Figure: Rooftop layout

4. Implementation

4.1 Description

Fittings :

i) Number of lights calculation

Number of lights

$$\sum NF = \frac{E \times A(\mathrm{m}^2)}{UF \cdot LLF} = [N]$$

Where
E = Required illuminance (lux)
A= Area of the room (m^2)
UF= Utilization Factor
LLF= Light Loss Factor
N= Number of luminaires

We have used UF. LLF =0.75 for our calculations.

ii) Number of fans calculation

Number of fans

$$M = \frac{A(ft^2)}{100}, Fans$$

Where, A= Room area in ft^2

The final number is selected to the nearest suitable integer based on room shape and comfort requirement.

Recommended Values of Illumination for Residential Area:

Floor type	Room type	E(lux)
Typical Floor	Dinning	150
	Drawing Room	100
	Kitchen	200
	Bedroom	100
	Veranda	70
	Reading Room	100
	Store Room	100
	Laundry	150
	Toilet	100
	Guest Bedroom	100

Fluorescent lamp luminous flux table:

Lamp type	Wattage (W)	Luminous flux (lm)
Ceiling Mounted light	16	950
LED bulb	20	1200
Tube Light	25	1500

The given tables introduce the values we used for the number of light fan calculations in each room. Variety and standard were ensured as per instructions. Some sample calculations are shown:

Master-Bed (15'-8"×13'-8")

UF × LLF = 0.75

$A = 19.89248152\ m^2,\ E = 100$

Number of light

$$\sum NF = \frac{E \times A(m^2)}{UF \cdot LLF} = \lceil N \rceil$$

$$\sum NF = \frac{100 \times 19.89248152}{0.75} = 2652.33$$

So, we will use 1 tube light and 2 LED bulbs

$$A = 214.120889\ ft^2$$

Number of fans

$$M = \frac{A(ft^2)}{100}, \text{Fans} \approx \text{round}(M)$$

$$M = \frac{214.120889}{100} = 2.14 \Rightarrow 2 \text{ fans}$$

Bedroom ((10'-0"×13'-8")

UF × LLF = 0.75

$$A = 12.69705848\ m^2,\ E = 100$$

$$\sum NF = \frac{100 \times 12.69705848}{0.75} = 1693.33$$

So, we will use 1 tube light and 1 LED bulbs

$$A = 136.67\ ft^2$$

$$M = \frac{136.67}{100} = 1.37 \Rightarrow 1 \text{ fan}$$

Bedroom (13'-9"×10'-8")

UF × LLF = 0.75

$$A = 13.62620501\ m^2,\ E = 100$$

$$\sum NF = \frac{100 \times 13.62620501}{0.75} = 1816.8$$

So, we will use 1 tube light and 1 LED bulbs

$$A = 146.67125\ ft^2$$

$$M = \frac{146.67125}{100} = 1.47 \Rightarrow 1 \text{ fan}$$

Bedroom (8'-3"×10'-8")

L=8’-3” =8.25 ft= 2.5146m

W=10’-8” =10.667 ft= 3.2512 m

UF × LLF = 0.75

$$A = 8.17546752\ m^2,\ E = 100$$

$$\sum NF = \frac{100 \times 8.17546752}{0.75} = 1090.062$$

So, we will use 1 LED bulb

$$A = 88.00275\ ft^2$$

$$M = \frac{88.00275}{100} = 0.8800275 \Rightarrow 1 \text{ fan}$$

Living + Dining (39'-2"×13'-0")

UF × LLF = 0.75

$$A = 47.30353378\ m^2,\ E = 150$$

$$\sum NF = \frac{150 \times 47.30353378}{0.75} = 9460.7$$

So, we will use 3 tube lights, 1 LED bulb and 4 ceiling mounted light

$$A = 509.171\ ft^2$$

$$M = \frac{509.171}{100} = 5.09 \Rightarrow 5 \text{ fans}$$

VER (9'×4')

UF × LLF = 0.75

$$A = 3.34450944\ m^2,\ E = 70$$

$$\sum NF = \frac{70 \times 3.34450944}{0.75} = 312.106$$

So, we will use 1 ceiling mounted light

VER (7'-4"×3'-8")

UF × LLF = 0.75

$$A = 2.497151031\ m^2,\ E = 70$$

$$\sum NF = \frac{70 \times 2.497151031}{0.75} = 233.07$$

So, we will use 1 ceiling mounted light

VER (13'-9"×2'-6")

UF × LLF = 0.75

$$A = 3.193542\ m^2,\ E = 70$$

$$\sum NF = \frac{70 \times 3.193542}{0.75} = 298.0639$$

So, we will use 1 ceiling mounted light

VER (8'-3"×2'-6")

L=8'-3" =8.25 ft= 2.5146m

W=2'-6" =2.5 ft= 0.762m

UF × LLF = 0.75

$$A = 1.9161252\ m^2,\ E = 70$$

$$\sum NF = \frac{70 \times 1.9161252}{0.75} = 178.838352$$

So, we will use 1 ceiling mounted light

Toilet (4'×8'-8")

UF × LLF = 0.75

$$A = 3.220762591\ m^2,\ E = 100$$

$$\sum NF = \frac{100 \times 3.220762591}{0.75} = 429.435$$

So, we will use 1 LED bulb

Toilet (4'×9')

L=8' =4 ft= 1.2192 m

W=9 =9 ft= 2.7432 m

UF × LLF = 0.75

$$A = 3.34450944\ m^2,\ E = 100$$

$$\sum NF = \frac{100 \times 3.34450944}{0.75} = 445.934592$$

So, we will use 1 LED bulb

Toilet (4'×7')

UF × LLF = 0.75

$$A = 2.60128512\ m^2,\ E = 100$$

$$\sum NF = \frac{100 \times 2.60128512}{0.75} = 346.838$$

So, we will use 1 LED bulb

Toilet (4'×6'-2")

UF × LLF = 0.75

$$A = 2.291620707\ m^2,\ E = 100$$

$$\sum NF = \frac{100 \times 2.291620707}{0.75} = 305.549$$

So, we will use 1 LED bulb

Kitchen (8'-0"×13'-8")

UF × LLF = 0.75

$$A = 10.1\ m^2,\ E = 200$$

$$\sum NF = \frac{200 \times 10.1}{0.75} = 2693.33$$

So, we will use 1 tube light and 1 LED bulb

$$A = 109.336\ ft^2$$

$$M = \frac{109.336}{100} = 1.09 \Rightarrow 1\ fan$$

By putting all the fans and lights according to the calculations in respective rooms, we get the following design.

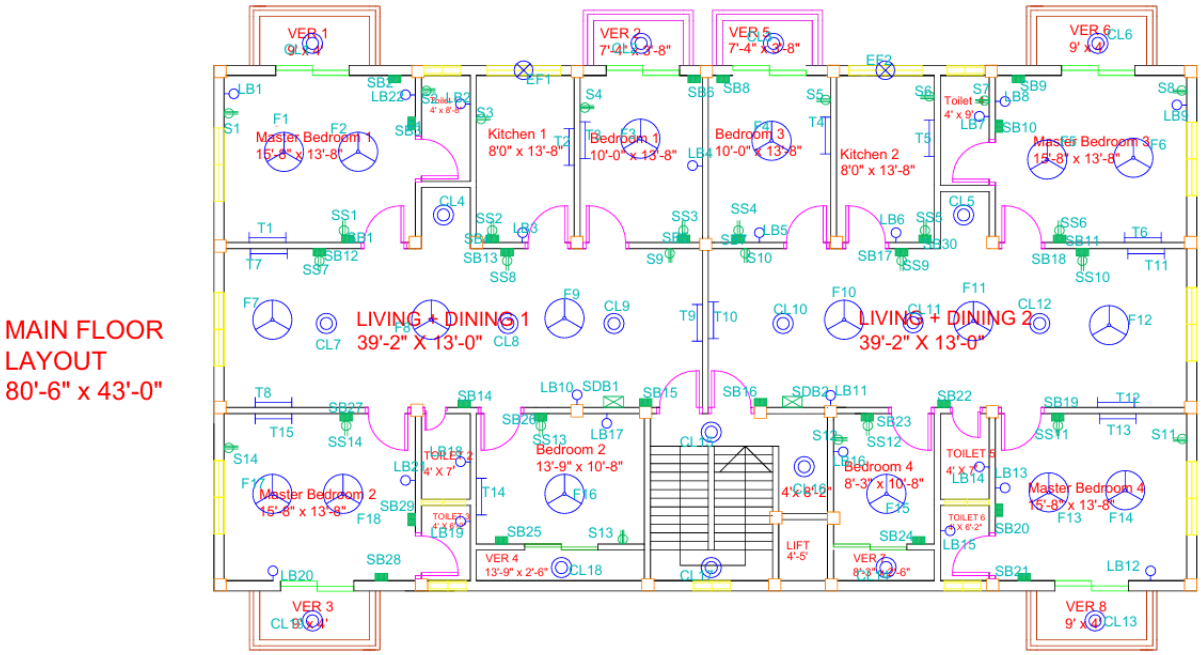


Figure: Main floor with fittings

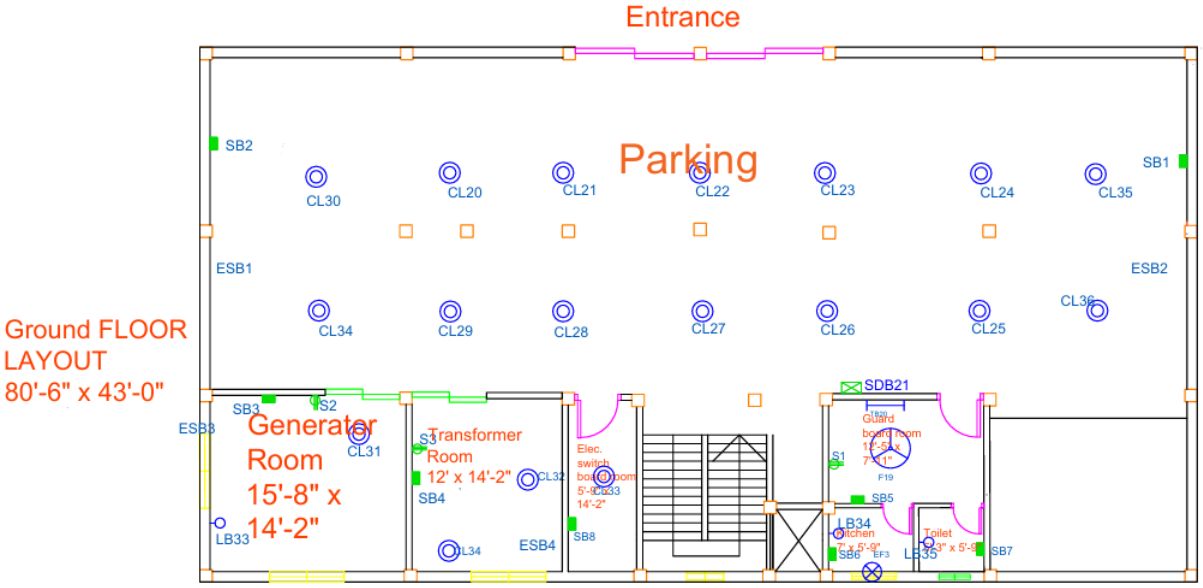


Figure: Ground Floor with fittings

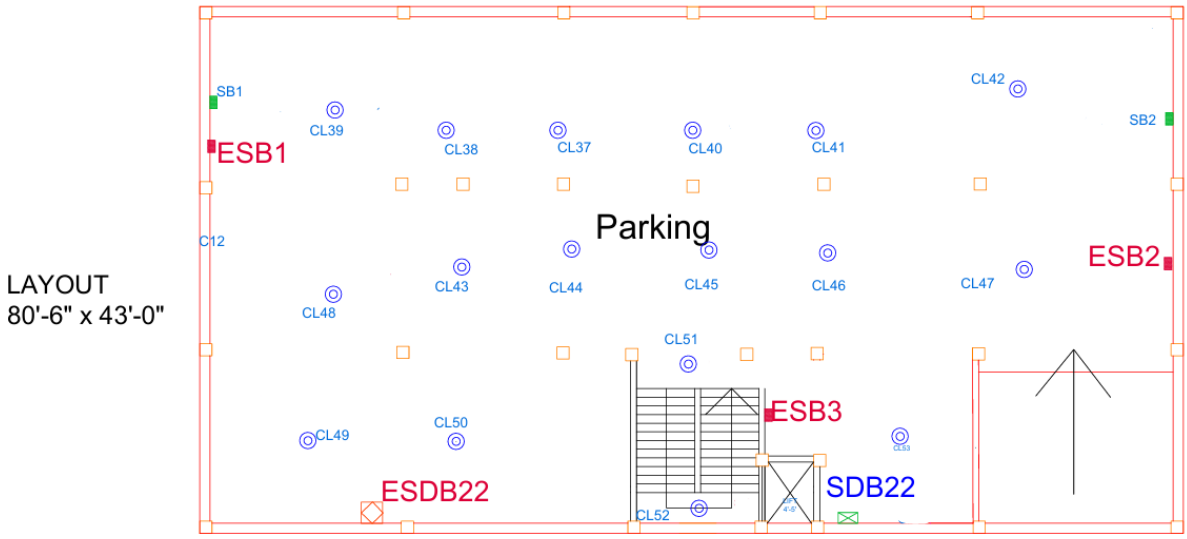


Figure: Basement with fittings

Conduit Planning:

We have done conduit planning according to the BNBC 2020 and done the following calculation.

symbol	Wire Rating (Single core)-mm^2	Current Rating (Ampere)	GI Pipe Diameter (inch)
C1	2 x 1.5 mm^2	5 A	3/4
C2	4 x 1.5 mm^2	5 A	3/4
C3	6 x 1.5 mm^2	5 A	3/4
C4	8 x 1.5 mm^2	5 A	3/4
C5	10 x 1.5 mm^2	5 A	1
C6	12 x 1.5 mm^2	5 A	1
C7	14 x 1.5 mm^2	5 A	1
C8	2 x 2.5 mm^2 +ECC	10 A	1
C9	2 x 4 mm^2 +ECC	15 A	1
C8,9	2 x 2.5 mm^2 and 2 x 4 mm^2 + ECC		1
C10	4 x 4 mm^2 +ECC	15 A	1
C11	2 x 2.5 mm^2 and 6 x 4 mm^2 + ECC		
C12	2 x 1.5 mm^2 +ECC	5 A	3/4
C9,12	2 x 1.5 mm^2 and 2 x 4 mm^2 + ECC		1

Calculations for current ratings:

Ceiling mounted light:

Power rating, P =16 W

Voltage, V =220 V

Power factor, pf = 0.9

Current flow through the ceiling mounted light,

$I=\frac{P}{V*pf}=\frac{16}{220*0.9}=0.0808\ A<5\ A$

So, 2 x 1.5 mm^2 BYM wire is needed

LED bulb:

Power rating, P =20 W

Voltage, V =220 V

Power factor, pf = 0.9

Current flow through the ceiling mounted light,

$I=\frac{P}{V*pf}=\frac{20}{220*0.9}=0.1010\ A<5\ A$

So, 2 x 1.5 mm^2 BYM wire is needed

Tube light:

Power rating, P =25 W

Voltage, V =220 V

Power factor, pf = 0.9

Current flow through the ceiling mounted light,

$I=\frac{P}{V*pf}=\frac{25}{220*0.9}=0.12626\ A<5\ A$

So, 2 x 1.5 mm^2 BYM wire is needed

Ceiling fan:

Power rating, P =75 W

Voltage, V =220 V

Power factor, pf = 0.8

Current flow through the ceiling mounted light,

$I=\frac{P}{V*pf}=\frac{75}{220*0.8}=0.426136\ A<5\ A$

So, 2 x 1.5 mm^2 BYM wire is needed

Exhaust fan:

Power rating, P =20 W

Voltage, V =220 V

Power factor, pf = 0.8

Current flow through the ceiling mounted light,

$I=\frac{P}{V*pf}=\frac{20}{220*0.8}=0.1136\ A<5\ A$

So, 2 x 1.5 mm^2 BYM wire is needed

2 pin sockets (SS):

Power rating, P =200 W

Voltage, V =220 V

Power factor, pf = 0.75

Current flow through the ceiling mounted light,

$I=\frac{P}{V*pf}=\frac{200}{220*0.75}=1.21\ A<5\ A$

So, 2 x 1.5 mm^2 BYM wire is needed

2 pin sockets for TV(ST):

Power rating, P =80 W (55inch LED TV)

Voltage, V =220 V

Power factor, pf = 0.75

Current flow through the ceiling mounted light,

$I=\frac{P}{V*pf}=\frac{80}{220*0.75}=0.4848\ A<5\ A$

So, 2 x 1.5 mm^2 BYM wire is needed

Power sockets:

Power rating, P =2000 W (Air Conditioner)

Voltage, V =220 V

Power factor, pf = 0.8

Current flow through the ceiling mounted light,

$I=\frac{P}{V*pf}=\frac{2000}{220*0.8}=11.36\ A<15\ A$

So, 2 x 4 mm^2 BYM + 4 mm^2 BYA ECC wire is needed

The floor conduit plan summary table follows the calculation of each load as described here:

Main conduit switchboards:

SDB1: (Typical floor unit 1)

Room name	Circuit no.	switchboard	Fixtures	Power(W)	Current rating	Wire rating	Breaker to SDB
Master bedroom 1	CKT1	SB1	T1	25	0.12626	C8	10 Ampere
			LB22	20	0.1010		
			F2	75	0.426136		
			SS1 (2 pin socket)		5		
		SB2	CL1	16	0.0808	C1	
		SB3	LB2	20	0.1010	C1	
				Total	5.835196		
Kitchen 1	CKT2	SB4	EF1	20	0.1136 A	C8	10 Ampere
			T2	25	0.12626		
			SS2(2 pin socket)		5		
				Total	5.23986		
Bedroom 1	CKT3	SB5	T3	25	0.12626	C8	10 Ampere
			F3	75	0.426136		
			SS3 (2 pin socket)		5		
		SB6	CL2	16	0.0808	C1	
				total	5.633196		
Master bedroom 2	CKT4	SB27	LB20	20	0.1010	C8	10 Ampere
			F17	75	0.426136		
			T15	25	0.12626		
			SS14 (2 pin socket)		5		
		SB28	CL19	16	0.0808	C1	
		SB29	LB19	20	0.1010	C1	
				Total	5.835196		
Bedroom 2	CKT5	SB25	CL18	16	0.0808	C1	10 Ampere
		SB26	T14	25	0.12626	C8	
			SS13(2 pin socket)		5		
				Total	5.20706		
Living + Dinning 1	CKT6	SB12	T7	25	0.12626	C8	10 Ampere
			T8	25	0.12626		
			F7	75	0.426136		
			CL7	16	0.0808		
			SS7 (2 pin socket)		5		
				Total	5.759456		
	CKT7	SB13	CL9	16	0.0808	C8	10 Ampere
			CL4	16	0.0808		
			F9	75	0.426136		
			SS8 (2 pin socket)		5		
				Total	5.587736		
	CKT8	SB14	LB18	20	0.1010	C1	5 Ampere
				Total	0.1010		
	CKT9	SB15	T9	25	0.12626	C1	5 Ampere
			LB10	20	0.1010		
				Total	0.22726		
Total Fixtures current ratings					39.42596		

Room name	Power socket	Current rating	Wire rating
Master bedroom 1	S1	15 A	C9
Toilet 1	S2	15 A	C9
Kitchen 1	S3	15 A	C9
Bedroom 1	S4	15 A	C9
Master bedroom 2	S14	15 A	C9
Bedroom 2	S13	15 A	C9
Living + Dinning 1	S9	15 A	C9

SDB2: (Typical floor unit 2)

Room name	Circuit no.	switchboard	Fixtures	Power(W)	Current rating	Wire rating	Breaker to SDB
Master bedroom 3	CKT10	SB9	CL6	16	0.0808	C1	10 Ampere
		SB10	LB7	20	0.1010	C1	
		SB11	LB8	20	0.1010	C8	
			F5	75	0.426136		
			SS6(2 pin socket)		5		
				Total	5.708936		
Kitchen 2	CKT11	SB30	EF2	20	0.1136 A	C8	10 Ampere
			T5	25	0.12626		
			SS5(2 pin socket)		5		
				Total	5.23986		
Bedroom 3	CKT12	SB7	T4	25	0.12626	C8	10 Ampere
			F4	75	0.426136		
			SS4 (2 pin socket)		5		
		SB8	CL3	16	0.0808	C1	
				total	5.633196		
Master bedroom 4	CKT13	SB19	LB13	20	0.1010	C8	10 Ampere
			F13	75	0.426136		
			T13	25	0.12626		
			SS11(2 pin socket)		5		
		SB20	LB15	20	0.1010	C1	
		SB21	CL13	16	0.0808	C1	
				Total	5.835196		
Bedroom 4	CKT14	SB24	CL14	16	0.0808	C1	10 Ampere
		SB23	LB16	20	0.1010	C8	
			SS12(2 pin socket)		5		
				Total	5.1818		
Living + Dinning 2	CKT15	SB16	T10	25	0.12626	C1	5 Ampere
				Total	0.12626		
	CKT16	SB17	CL5	16	0.0808		10 Ampere
			CL10	16	0.0808	C8	
			F10	75	0.426136		
			LB11	20	0.1010		
			SS9 (2 pin socket)		5		
				Total	5.688736		
	CKT17	SB18	F12	75	0.426136	C8	10 Ampere
			CL12	16	0.0808		
			T11	25	0.12626		
			T12	25	0.12626		
			SS10 (2 pin socket)		5		
				Total	5.759456		
	CKT18	SB22	LB14	20	0.1010	C1	5 Ampere
				Total	0.1010		
Total Fixtures current ratings					39.27444		

Room name	Power socket	Current rating	Wire rating
Master bedroom 3	S8	15 A	C9
Toilet 4	S7	15 A	C9
Kitchen 2	S6	15 A	C9
Bedroom 3	S5	15 A	C9
Master bedroom 4	S11	15 A	C9
Bedroom 4	S12	15 A	C9
Living + Dinning 2	S10	15 A	C9

SDB21: (Ground floor)

Room name	Circuit no.	switchboard	Fixtures	Power(W)	Current rating	Wire rating	Breaker to SDB
Parking	CKT16	SB1	CL22	16	0.0808	C1	5 Ampere
			CL23	16	0.0808		
			CL24	16	0.0808		
			CL35	16	0.0808		
				Total	0.3232		
	CKT17	SB2	CL20	16	0.0808	C1	5 Ampere
			CL21	16	0.0808		
			CL30	16	0.0808		
				Total	0.2424		
Generator room	CKT18	SB3	LB33	20	0.1010	C1	5 Ampere
				total	0.1010		
Transformer room	CKT19	SB4	CL34	16	0.0808	C1	5 Ampere
				Total	0.0808		
Elec. Switchboard room	CKT20	SB8	CL33	16	0.0808	C1	5 Ampere
				Total	0.0808		
Kitchen	CK21	SB6	EF3	20	0.1010	C1	5 Ampere
			LB34	20	0.1010		
				Total	0.2020		
Guard room	CK22	SB5	F19	75	0.426136	C1	5 Ampere
			TB20	25	0.12626		
				Total	0.552396		
Toilet	CKT23	SB7	LB35	20	0.1010	C1	5 Ampere

Room name	Power socket	Current rating	Wire rating
Generator room	S2	15 A	C9
Transformer room	S3	15 A	C9
Guard room	S1	15 A	C9

SDB22: (Basement)

Room name	Circuit no.	switchboard	Fixtures	Power(W)	Current rating	Wire rating	Breaker to SDB
Parking	CKT28	SB1	CL37	16	0.0808	C1	5 Ampere
			CL38	16	0.0808		
			CL39	16	0.0808		
				Total	0.2424		
	CKT29	SB2	CL40	16	0.0808	C1	5 Ampere
			CL41	16	0.0808		
			CL42	16	0.0808		
				Total	0.2424		

We also added emergency switchboards in the typical floor, basement, and ground floor.

ESDB1:

Room Name	Circuit No	Switchboard	Fixtures	Power (W)	Current Rating	Wire Rating	Breaker to ESDB
M. Bedroom 1	CKT1	ESB1	LB1	20	0.12987013	C1	5 Ampere
			F1	75	0.487012987		
			Total		0.616883117		
Bedroom 1	CKT3	ESB4	LB4	20	0.12987013	C1	5 Ampere
			Total		0.12987013		
M. Bedroom 2	CKT4	ESB5	LB21	20	0.12987013	C1	5 Ampere
			F18	75	0.487012987		
			Total		0.616883117		
Bedroom 2	CKT5	ESB6	F16	75	0.487012987	C1	5 Ampere
			Total		0.487012987		
Kitchen 1	CKT2	ESB2	LB3	20	0.12987013	C1	5 Ampere
			Total		0.12987013		
Living + Dinning 1	CKT6	ESB3	F8	75	0.487012987	C1	5 Ampere
			CL8	16	0.103896104		
			Total		0.590909091		
			Total Fixtures Current Rating		2.571428572		

ESDB2:

Room Name	Circuit No	Switchboard	Fixtures	Power (W)	Current Rating	Wire Rating	Breaker to ESDB
M. Bedroom 3	CKT10	ESB9	LB9	20	0.12987013	C1	5 Ampere
			F6	75	0.487012987		
			Total		0.616883117		
Bedroom 3	CKT12	ESB7	LB5	20	0.12987013	C1	5 Ampere
			Total		0.12987013		
M. Bedroom 4	CKT13	ESB12	LB12	20	0.12987013	C1	5 Ampere
			F14	75	0.487012987		
			Total		0.616883117		
Bedroom 4	CKT14	ESB11	F15	75	0.487012987	C1	5 Ampere
			Total		0.487012987		
Kitchen 2	CKT11	ESB8	LB6	20	0.12987013	C1	5 Ampere
			Total		0.12987013		
Living + Dinning 2	CKT15	ESB10	F11	75	0.487012987	C1	5 Ampere
			CL11	16	0.103896104		
			Total		0.590909091		
			Total Fixtures Current Rating		2.571428572		

ESDB21: (Ground Floor)

Room name	Circuit no.	switchboard	Fixtures	Power(W)	Current rating	Wire rating	Breaker to SDB
Parking	CKT24	ESB1	CL28	16	0.0808	C1	5 Ampere
			CL29	16	0.0808		
			CL34	16	0.0808		
				Total	0.2424		
	CKT25	ESB2	CL25	16	0.0808	C1	5 Ampere
			CL26	16	0.0808		
			CL27	16	0.0808		
			CL36	16	0.0808		
				Total	0.3232		
Generator room	CKT26	ESB3	CL31	16	0.0808	C1	5 Ampere
				total	0.0808		
Transformer room	CKT27	ESB4	CL32	16	0.0808	C1	5 Ampere
				Total	0.0808		

ESDB22: (Basement)

Room name	Circuit no.	switchboard	Fixtures	Power(W)	Current rating	Wire rating	Breaker to SDB
Parking	CKT30	ESB1	CL43	16	0.0808	C1	5 Ampere
			CL44	16	0.0808		
			CL48	16	0.0808		
			CL50	16	0.0808		
			CL49	16	0.0808		
				Total	0.404		
	CKT31	ESB2	CL45	16	0.0808	C1	5 Ampere
			CL46	16	0.0808		
			CL47	16	0.0808		
			CL53	16	0.0808		
				Total	0.3232		
	CKT32	ESB3	CL51	16	0.0808	C1	5 Ampere
			CL52	16	0.0808		
				Total	0.1616		

After all necessary planning and calculations, we draw the conduits in AUTOCAD.

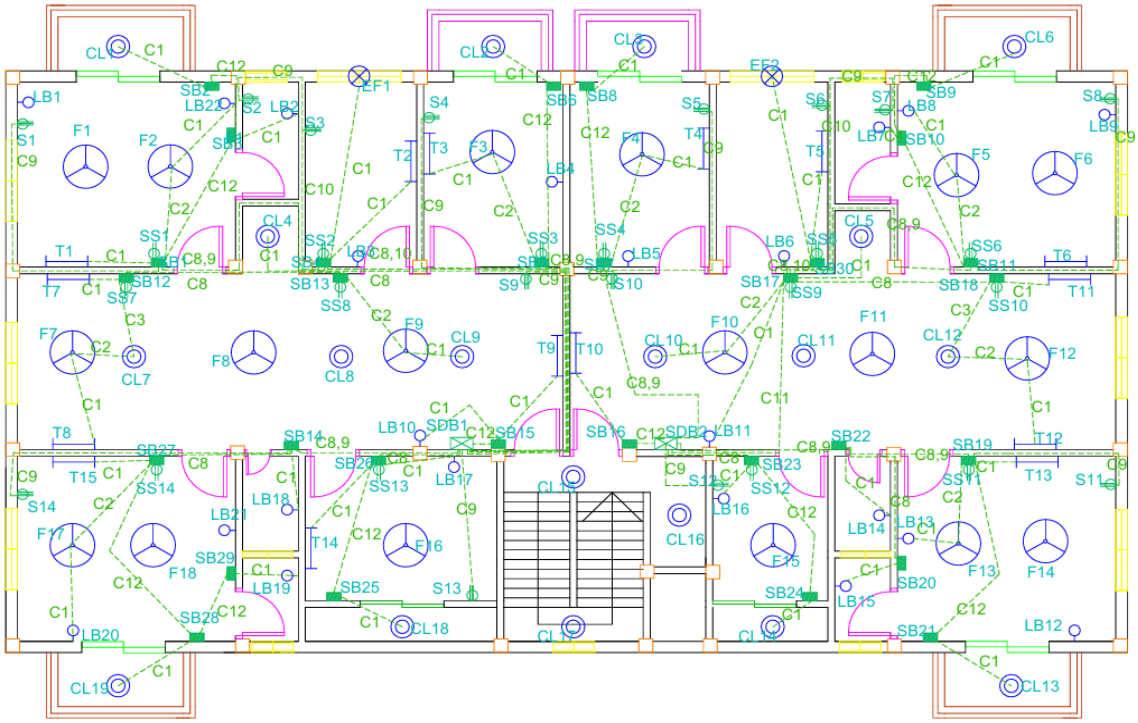


Figure : Typical floor conduit (main)

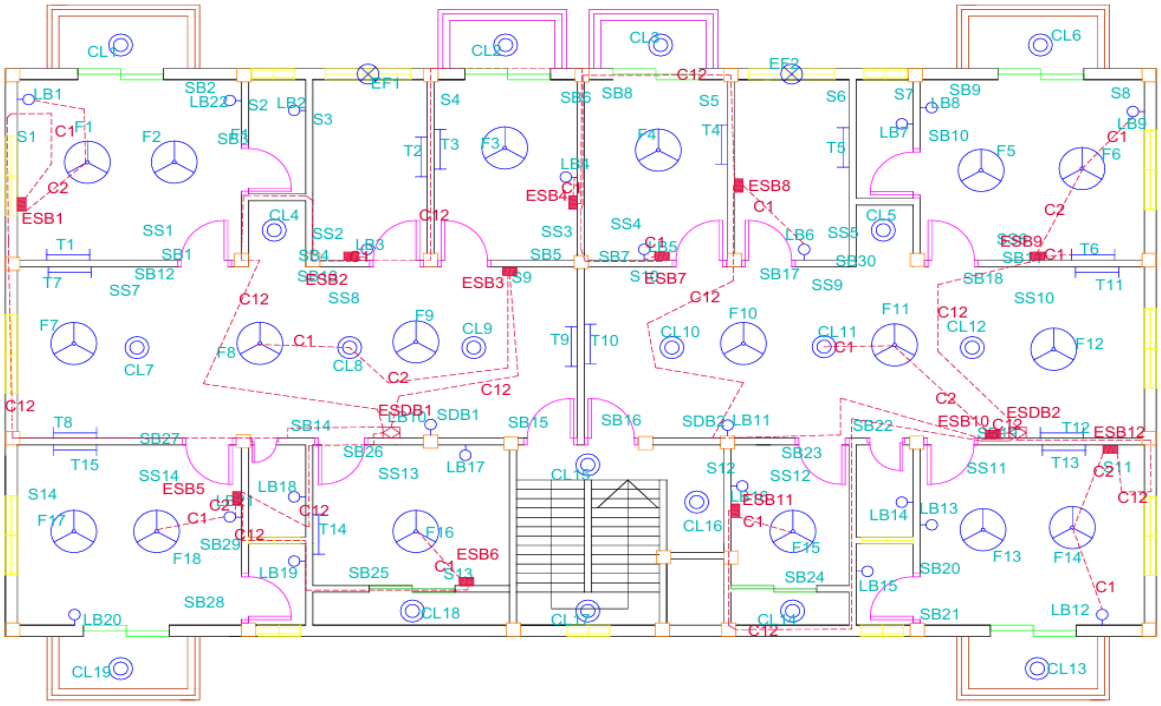


Figure : Typical floor conduit (emergency)

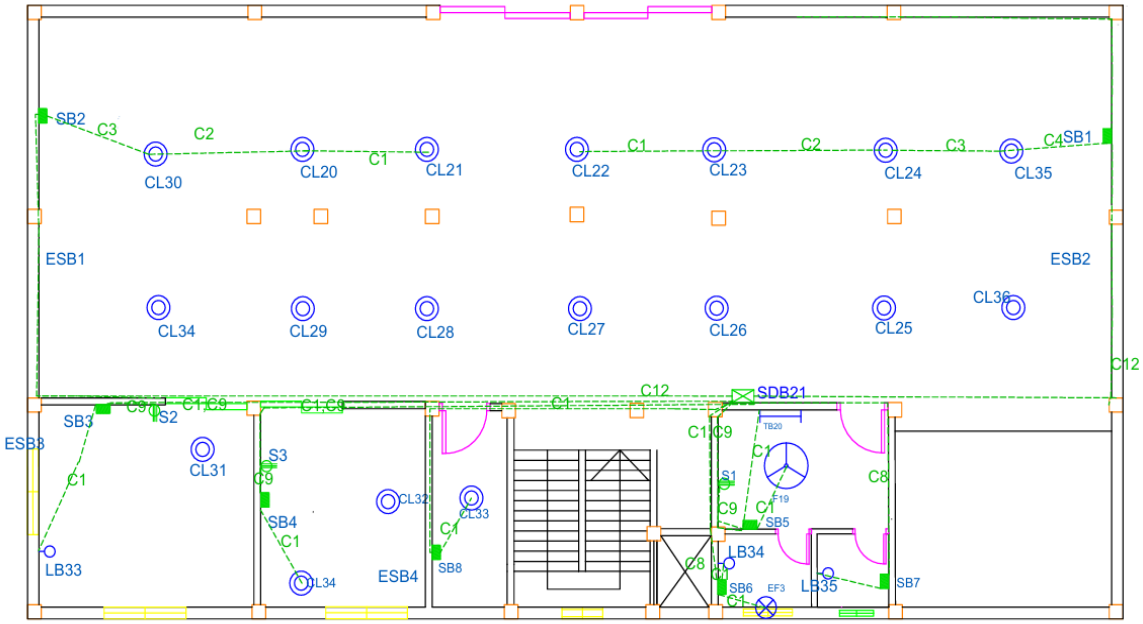


Figure: Ground floor conduit (main)

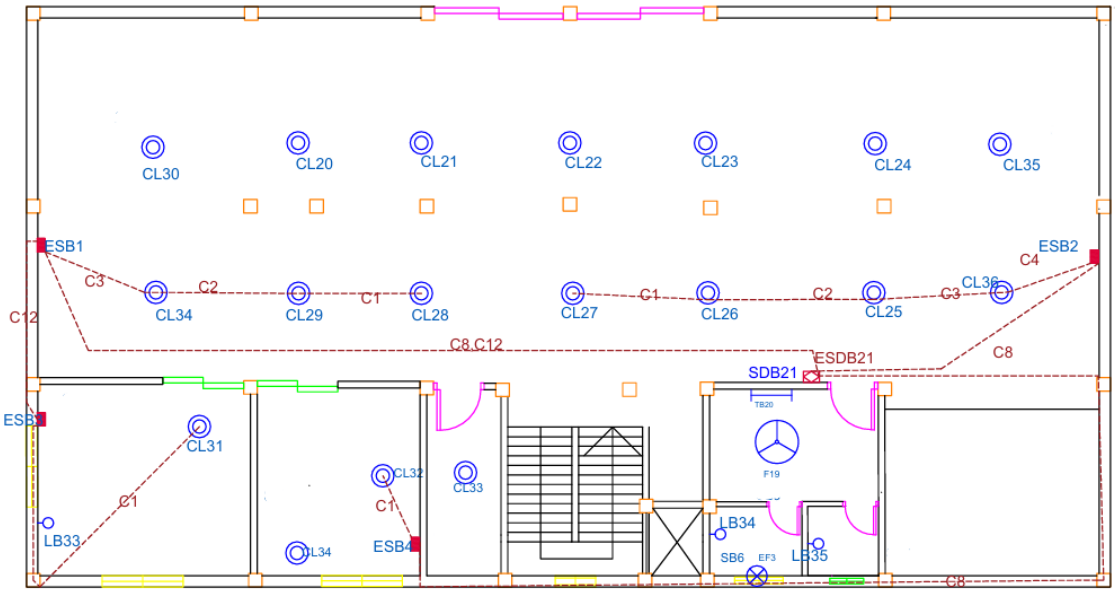


Figure: Ground floor conduit at emergency

LAYOUT
80'-6" x 43'-0"

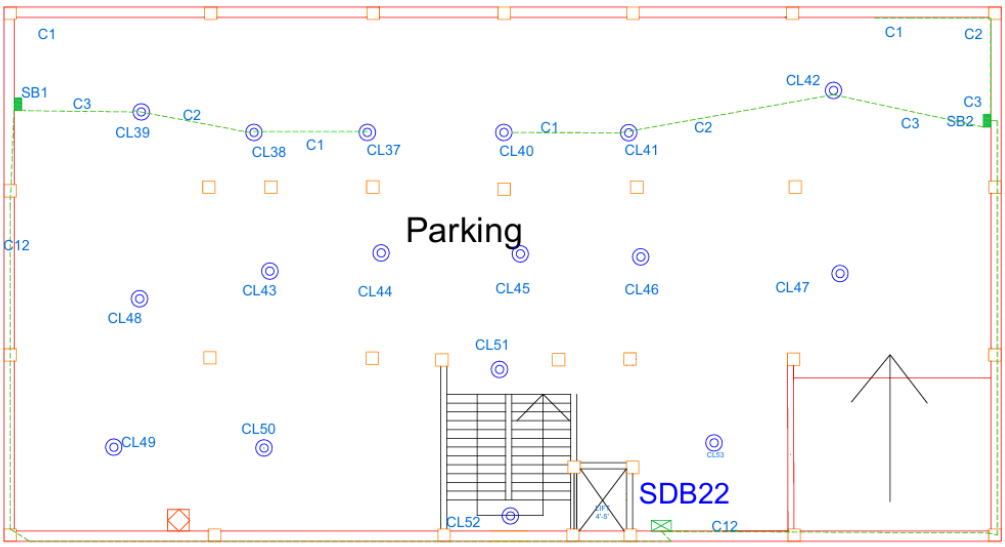


Figure: Basement conduit (main)

LAYOUT
80'-6" x 43'-0"

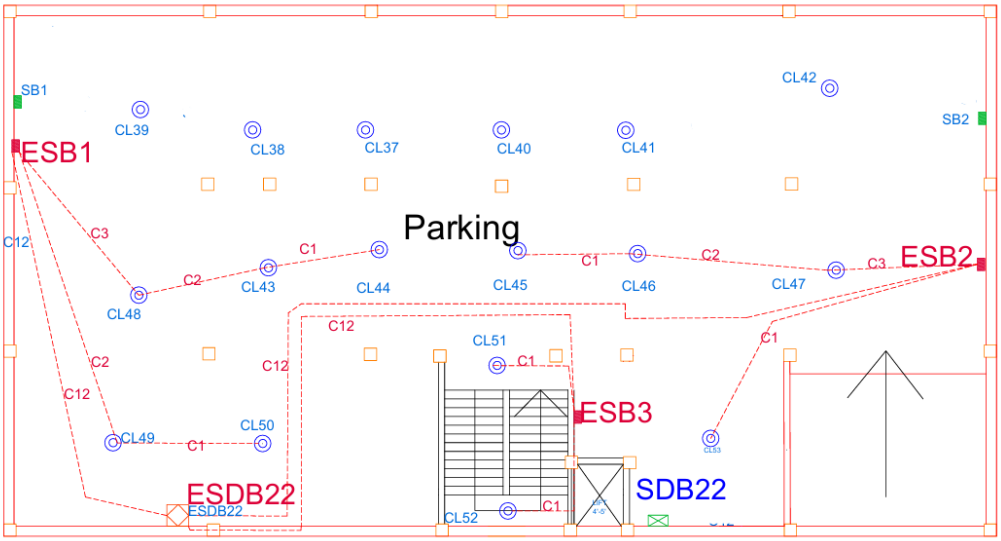


Figure: Basement conduit at emergency

Lightning Protection System

Length $= 80' - 6'' = 80.5$ ft = 24.54 m

Width $= 43$ ft = 13.11 m

Total area of rooftop, $A = 24.54 \times 13.11 = 321.78\ \text{m}^2$

Calculation for Number of Air Terminals (Rolling Sphere Method):

Let, floor-to-floor height $= 4$ m

The building is **10 storied**, then total building height is:

$$H = 10 \times 4 = 40\ \text{m}$$

For this design, **Level I protection** is selected.

The corresponding rolling sphere radii are:

Level I → 20 m

Level II → 30 m

Level III → 45 m

Level IV → 60 m

Since Level I is used, the rolling sphere radius is:

$$r = 20 \text{ m}$$

Let the height of each air terminal rod be:

$$h = 2 \text{ m}$$

For Level I protection, the maximum allowable distance between two adjacent air terminals is:

$$d_{\max} = 2\sqrt{2rh - h^2}$$
$$d_{\max} = 2\sqrt{2(20)(2) - (2)^2}$$
$$d_{\max} = 2\sqrt{80 - 4}$$
$$d_{\max} = 2\sqrt{76}$$
$$d_{\max} \approx 2 \times 8.718$$
$$d_{\max} \approx 17.43 \text{ m}$$

On the 24.54 m side:

Number of required segments = $\frac{24.54}{17.43} \approx 1.41$

Since this is greater than 1, we take 2 segments.

So, the number of air terminals needed along one 24.54 m side is = $2 + 1 = 3$

On the 13.11 m side:

Number of required segments = $\frac{13.11}{17.43} \approx 0.75$

So, we take 1 segment.

That means the number of air terminals needed along one 13.11 m side is = $1 + 1 = 2$

The roof has:

Two sides of 24.54 m, each requiring 3 air terminals

Two sides of 13.11 m, each requiring 2 air terminals

At first,

$$(2 \times 3) + (2 \times 2) = 6 + 4 = 10$$

But the corner air terminals are counted twice in this total. Since there are 4 corners, subtract 4 = $10 - 4 = 6$

Therefore, the total number of air terminals required is 6

Calculation for Number of Down Conductors

No. of down conductors, $n = 1 + \frac{(L \times W) - 80}{100}$

$$= 1 + \frac{(24.54 \times 13.11) - 80}{100}$$
$$= 3.4178$$

So, **4 down conductors** are required.

And lastly , we have added **roof conductor** peripherally to the building .

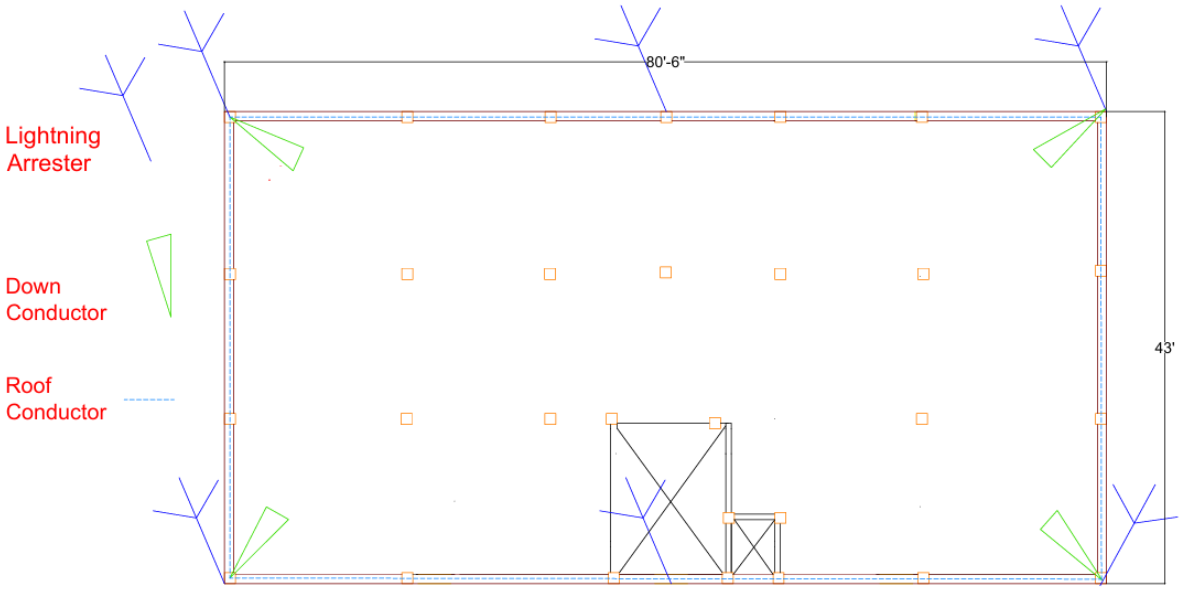


Figure: Rooftop lightning protection system

5. Design Analysis and Evaluation

5.1 Design Considerations (PO(c))

5.1.1 Considerations to public health and safety

This design ensures safety by following BNBC 2020 standards. A Level I lightning protection system using the rolling sphere method protects the building from natural hazards. An emergency power system with a 25 kW generator and ESDBs on each floor keeps critical loads (lift, lighting, fire systems) running during outages, ensuring safe operation and evacuation.

5.2.2 Considerations to environment

The design follows BNBC 2020 to ensure safety and environmental efficiency. It includes a Level I lightning protection system, proper grounding (ECC), and circuit protection (MCCBs). A backup generator and ESDBs ensure reliability. Energy efficiency is improved using a 174 kVAR PFI plant and LED lighting, reducing power loss and overall energy consumption.

5.2.3 Considerations to cultural and societal needs

The design supports safe and comfortable residential living. Safety is ensured through lightning protection and proper electrical sizing. Illumination levels (150 lux for dining, 100 lux for living/bedrooms) meet daily lifestyle needs. Backup systems (generator and ESDBs) maintain essential services like lifts and security, promoting a reliable community environment.

5.2 Investigations (PO(d))

5.2.1 Results and Analysis

Single Line Diagram

Following our design, we have drawn the Single Line Diagram(SLD) for all types of DB(Distribution Board) , the connection of SDB to MDB, and ESDB to EMDB. And lastly, drew the whole MDB and EMDB.

Each of the wiring specifications is related to certain calculations, which are shown below:

Total load calculations:

Board Fixtures	Power (W)	Power Circuits	Total (W)
SDB1	39.43 × 154 = 6,072	57.5 × 198 = 11,385	17,457
SDB2	39.27 × 154 = 6,048	57.5 × 198 = 11,385	17,433
ESDB1	2.57 × 154 = 396	0	396
ESDB2	2.57 × 154 = 396	0	396
SDB21	1.44 × 154 = 222	27.5 × 198 = 5,445	5,667

ESDB21	0.73 × 154 = 112	0	112
SDB22	0.48 × 154 = 74	0	74
ESDB22	0.89 × 154 = 137	0	137

Normal Load (SDBs):

1. 10 floors: 10 × (SDB1 + SDB2) = 10 × (17,457 + 17,433) = 10 × 34,890 = 348,900 W

    a. SDB21 = 5,667 W
    b. SDB22 = 74 W

Total Normal Load = 354,641 W ≈ 355 kW

Emergency Load (ESDBs):

2. Floors: 10 × (ESDB1 + ESDB2) = 10 × (396 + 396) = 10 × 792 = 7,920 W

    a. ESDB21 = 112 W
    b. ESDB22 = 137 W

Total Emergency Load = 8,169 W ≈ 8.2 kW

Additional Loads:

a. Lift = 18,500 W
b. Pump = 5,000 W

GRAND TOTAL = 354,641 + 8,169 + 18,500 + 5,000 = 386,310 W ≈ 386 kW

Considering a diversity factor of 0.7 is necessary.

Board	Total Load	After Diversity (×0.7)
MDB (Normal + Pump)	354,641 + 5,000 = 359,641 W	251,749 W
EMDB (Emergency + Lift)	8,169 + 18,500 = 26,669 W	18,668 W

MDB:

a. Per phase power = 251,749 ÷ 3 = 83,916 W
b. Per phase Maximum current = 83,916 ÷ (220 × 0.9) = 424 A

EMDB:

a. Per phase power = 18,668 ÷ 3 = 6,223 W
b. Per phase current = 6,223 ÷ (220 × 0.95) = 30 A

Transformer Size

a. Total load before diversity = 386,310 W
b. Transformer kVA = 386,310 ÷ (0.95 × 0.8) = 386,310 ÷ 0.76 = 508 kVA

Use 500 kVA or 630 kVA transformer

PFI Plant

MDB load after diversity = 251,749 W

Initial PF = 0.7 → tan θ1 = 1.02, Target PF = 0.95 → tan θ2 = 0.329

Required VAR = 251,749 × (1.02 - 0.329) = 251,749 × 0.691 = 174,000 VAR (174 kVAR)

Per phase VAR = 174,000 ÷ 3 = 58,000 VAR

VAR Current = 58,000 ÷ (220 × 0.312) = 58,000 ÷ 68.64 = 845 A

PFI Breaker = 1,000 A TP MCCB

Main and Emergency Distribution Board:

Emergency Sub Distribution Load:

Total number of ESDB =22

Total current for ESDB-1 to EMDB =2.57 A

ESDB per phase =7, 7, 8

Total ESDB load =8,169 W

Lift load =18,500 W

Total load on EMDB (diversity 0.7) =18,668 W

Total load per phase (EMDB)= 6,223 W

Lift breaker rating =30 A

Max current per phase (EMDB) =30 A

Sub Distribution Board:

Total number of SDB =22

Total current for SDB-1 to MDB (after diversity) =96.93 A

SDB per phase =7, 7, 8

Total SDB load (after diversity) =251,749 W

Pump load= 5,000 W

Total load on MDB (diversity already applied) =251,749 W

Total load per phase (MDB) =83,916 W

Pump breaker rating =15 A

Max current per phase (MDB Bas Bar 0.9 pf) =424 A

Transformer to MDB breaker : 630 A TP MCCB

Transformer line: 4 × 300 mm² NYY + 150 mm² NYY ECC

Power Meter Line:

Current to SDB1 96.93 A

Current to ESDB1 2.57 A

Total current for each stranded unit 99.5 A

Wire rating from Meter to bus bar wire:2 × 35 mm² BYA + 16 mm² BYA ECC

Generator Bus bar:

Generator current per phase 30 A

Generator rating = (3*phase voltage*Line Current)/pf = 25 kW (pf=0.8)

Generator breaker: 60 A TP MCCB

Generator line: 4 × 16 mm² NYY + 10 mm² NYY ECC

Transformer Calculation:

Total Load (SDB+ESDB+Pump+Lift) =386,310 W

Transformer kVA = Total Load /pf*safety factor =386,310 / 0.95 * 0.8 =500 KVA

Transformer required.

PFI Plant:

MDB Load Calculation

- MDB Load (P) 251,749 W

- Initial PF 0.7
- Target PF0.95

Reactive Power Compensation:

Required VAR=Q=Q1-Q2=P*[tan(theta1)- tan(theta2)] =174,000 VAR

Per phase VAR = Required VAR /3 =58,000 VAR

VAR Current I= Per phase VAR /V* sin(theta1)=845 A

PFI Breaker= 1,000 A TP MCCB

PFI Plant to Main Bus bar=4 × 500 mm² NYY + 240 mm² NYY ECC

Reference:

SCHEDULE OF CONDUIT

SYMBOL	CONTAING POWER CABLE + ECC	CONDUIT SIZE
C1	2 X 1.5mm² BYM + 1.5mm² BYA	16mm
C2	4 X 1.5mm² BYM + 1.5mm² BYA	20mm
C3	6 X 1.5mm² BYM + 1.5mm² BYA	25mm
C4	8 X 1.5mm² BYM + 1.5mm² BYA	30mm
C5	10 X 1.5mm² BYM + 1.5mm² BYA	30mm
C6	2 X 2.5mm² BYM + 2.5mm² BYA	16mm
C7	4 X 2.5mm² BYM + 2.5mm² BYA	30mm
C8	2 X 4mm² BYM + 4mm² BYA	20mm
C9	4 X 4mm² BYM + 4mm² BYA	25mm
C10	2 X 6mm² BYM + 6mm² BYA	25mm
C11	4 X 6mm² BYM + 6mm² BYA	30mm
C12	2 x 10mm² BYM + 10mm² BYA	25mm
C13	2 X 16mm² BYM + 16mm² BYA	30mm
C14	2 X 25mm² BYM + 25mm² BYA	40mm
C15	2 X 35mm² BYM + 16mm² BYA	40mm
C16	4 X 35mm² NYY + 16mm² BYA	50mm
C17	4 X 70mm² NYY + 1X35mm² BYA	65mm(GI PIPE
C18	2 x 1.5mm² BYM + 4mm² BYA	25mm
	2 X 4mm² BYM + 4mm² BYA	25mm
C28	4 X 1.5mm² BYM + 4mm² BYA	25mm
	2 X 4mm² BYM + 4mm² BYA	25mm
C38	6 X 1.5mm² BYM + 4mm² BYA	30mm
	2 X 4mm² BYM + 4mm² BYA	30mm

Following all these, single line diagrams from each floor to the supply grid are shown below:

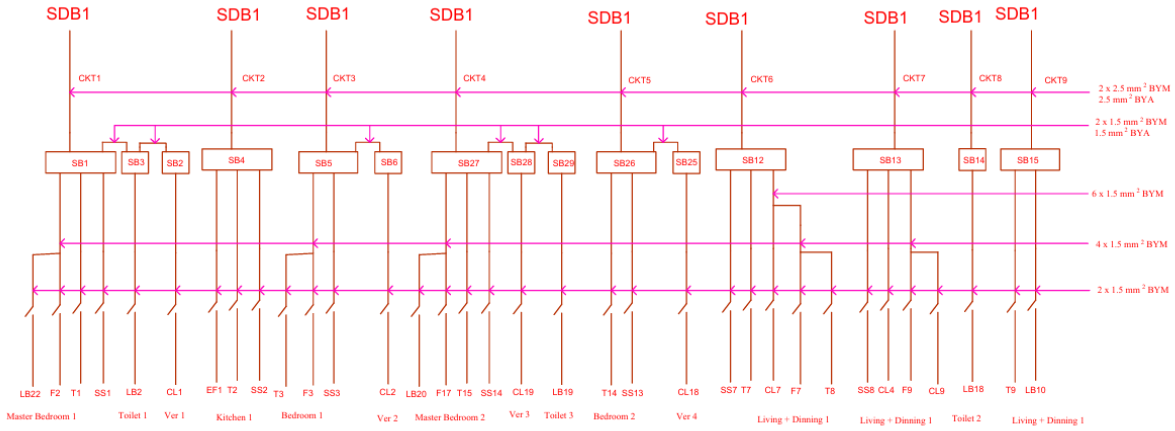


Figure : Switchboard diagram (Unit 1)

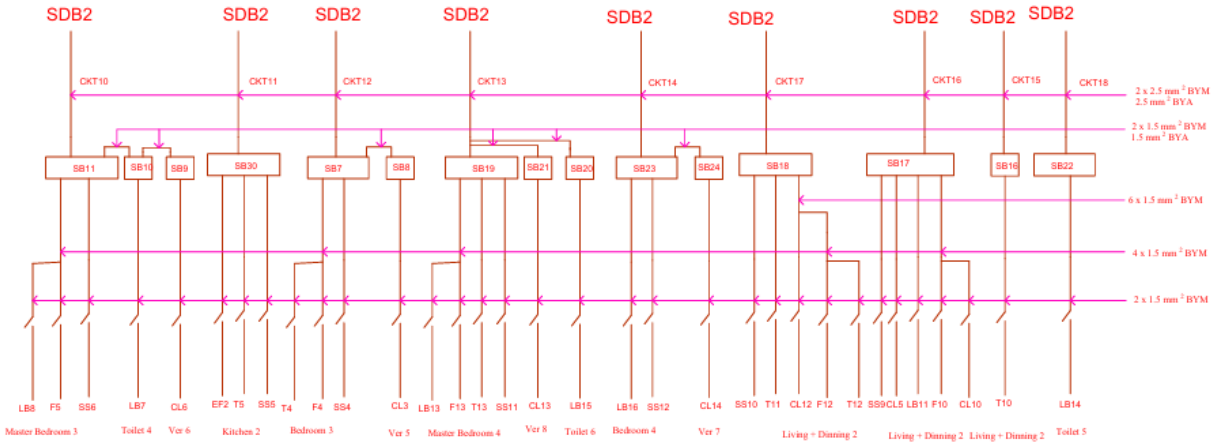


Figure: Switchboard diagram (Unit 2)

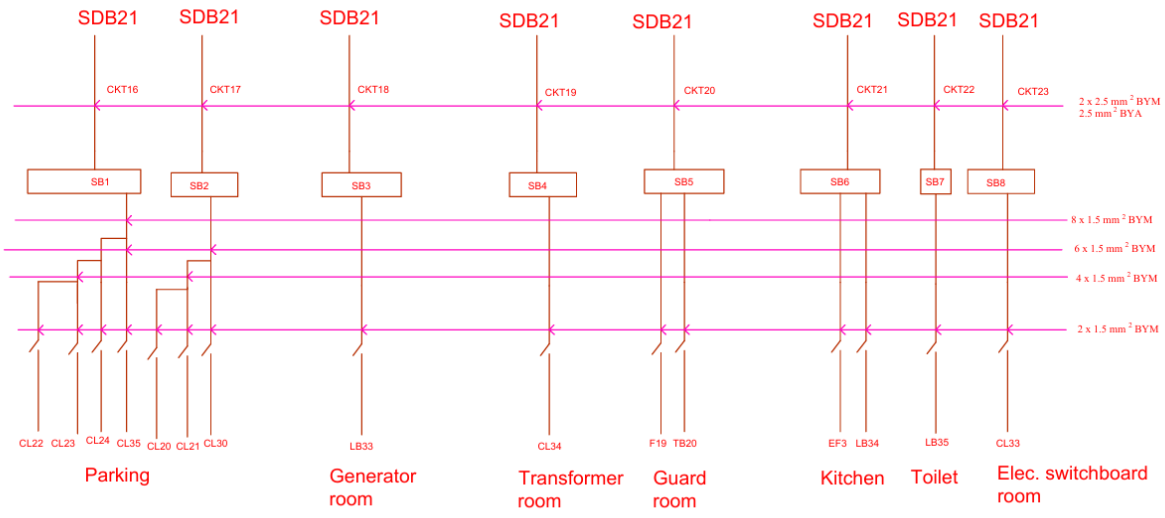


Figure: Switchboard Diagram (Ground Floor)

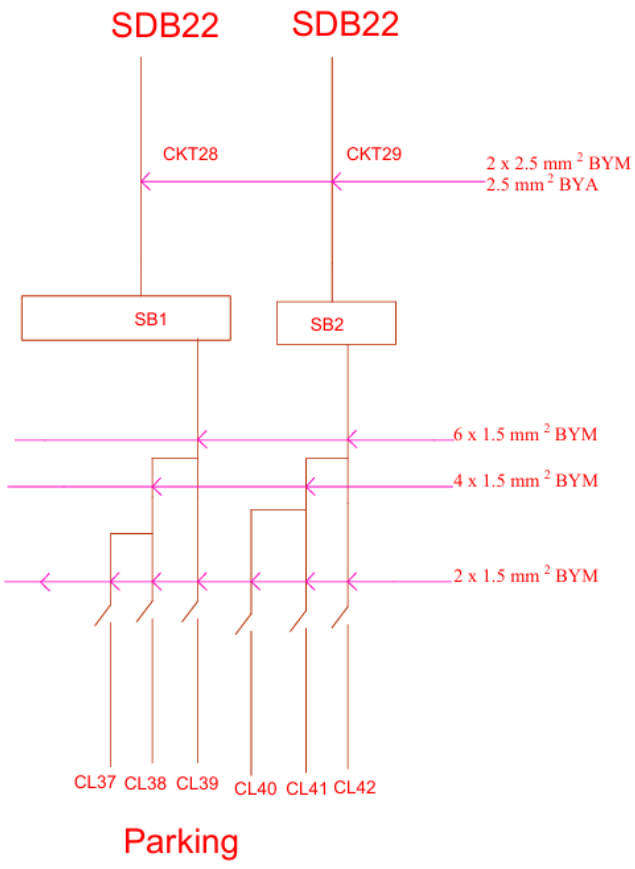


Figure: Switchboard Diagram (Basement)

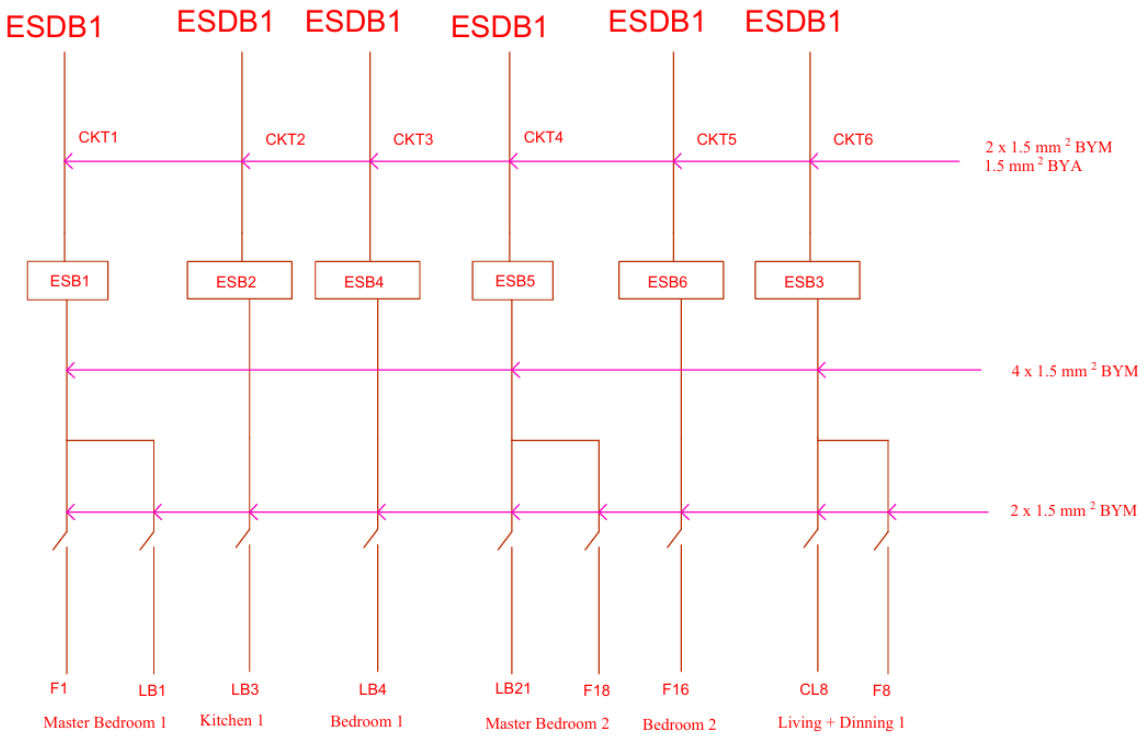


Figure: Emergency Switchboard Diagram (Unit 1)

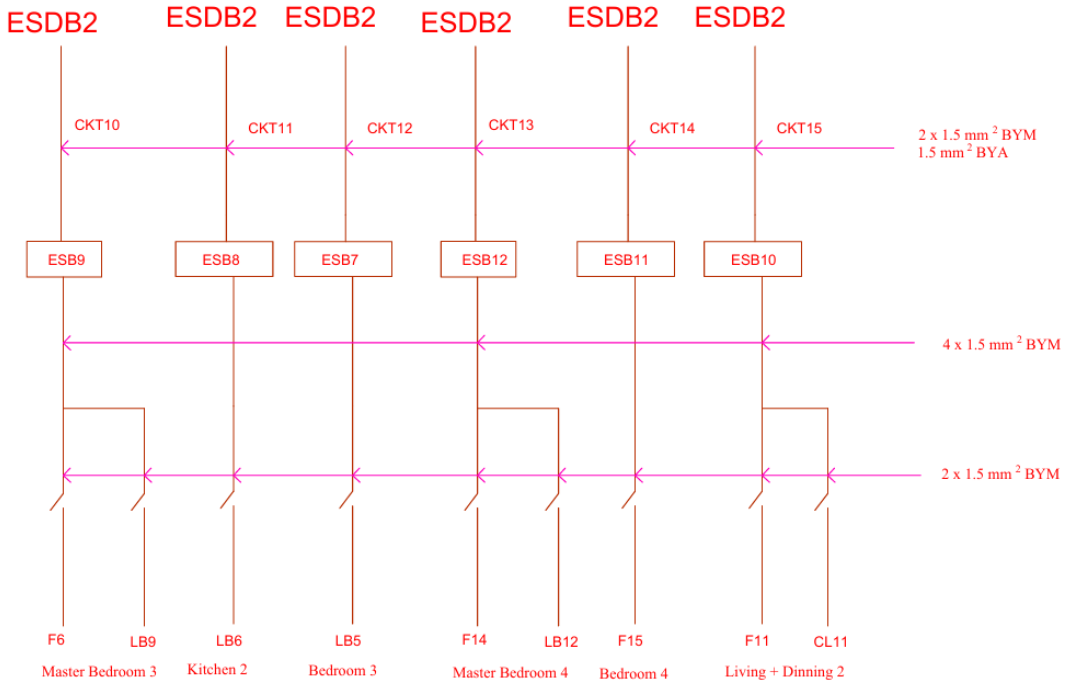


Figure: Emergency Switchboard Diagram (Unit 2)

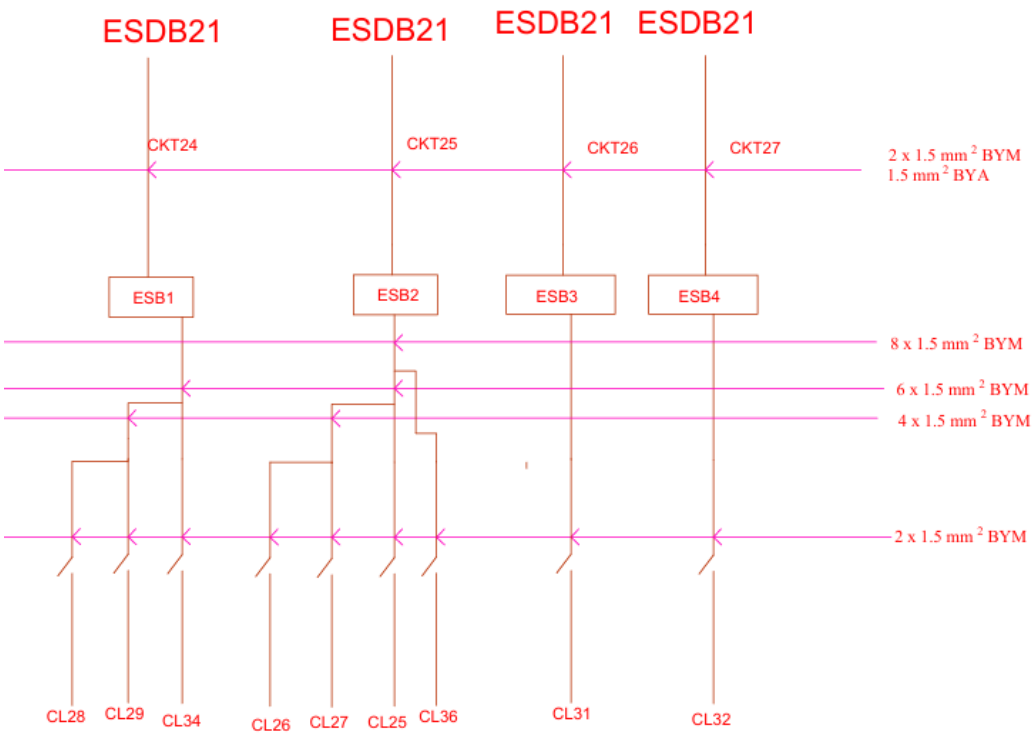


Figure: Emergency Switchboard diagram (Ground)

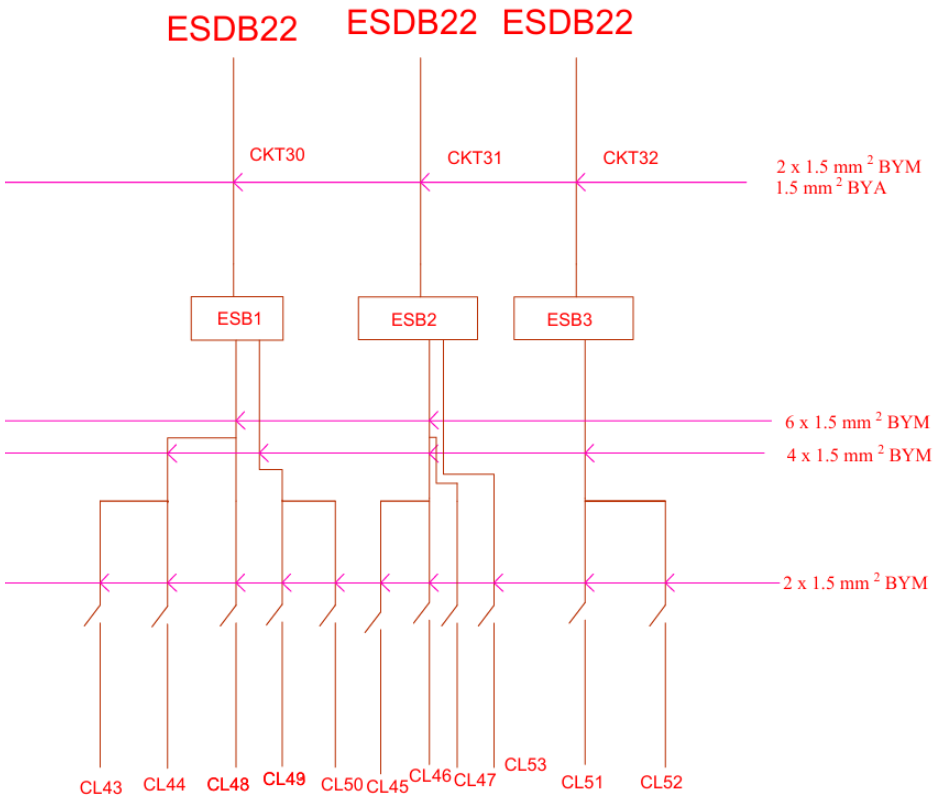


Figure: Emergency Switchboard diagram (Basement)

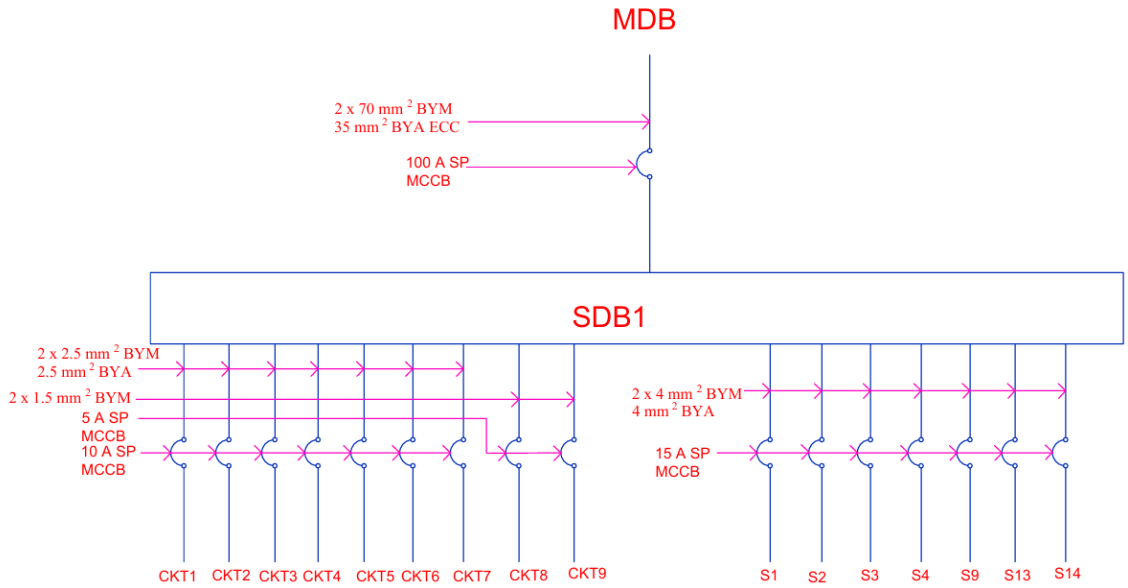


Figure: SDB Connection Diagram (Floor)

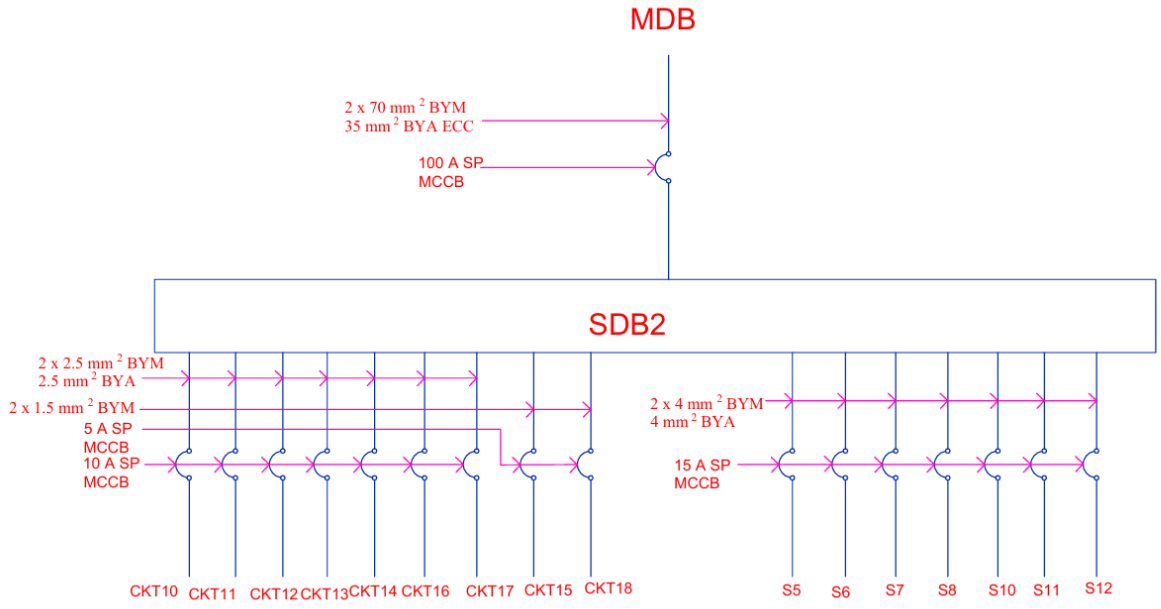


Figure: SDB Connection Diagram (Floor)

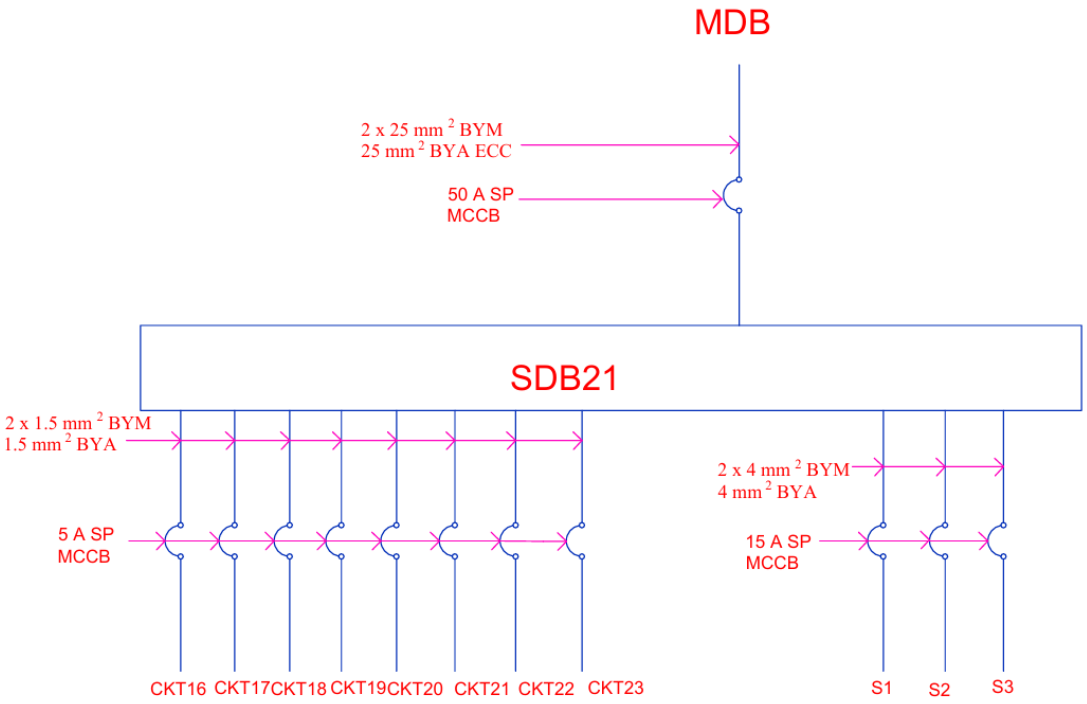


Figure: SDB Connection diagram (Ground)

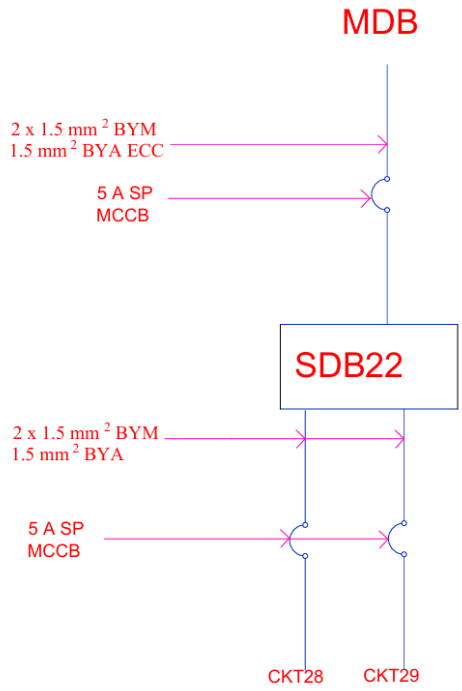


Figure: SDB Connection diagram (Basement)

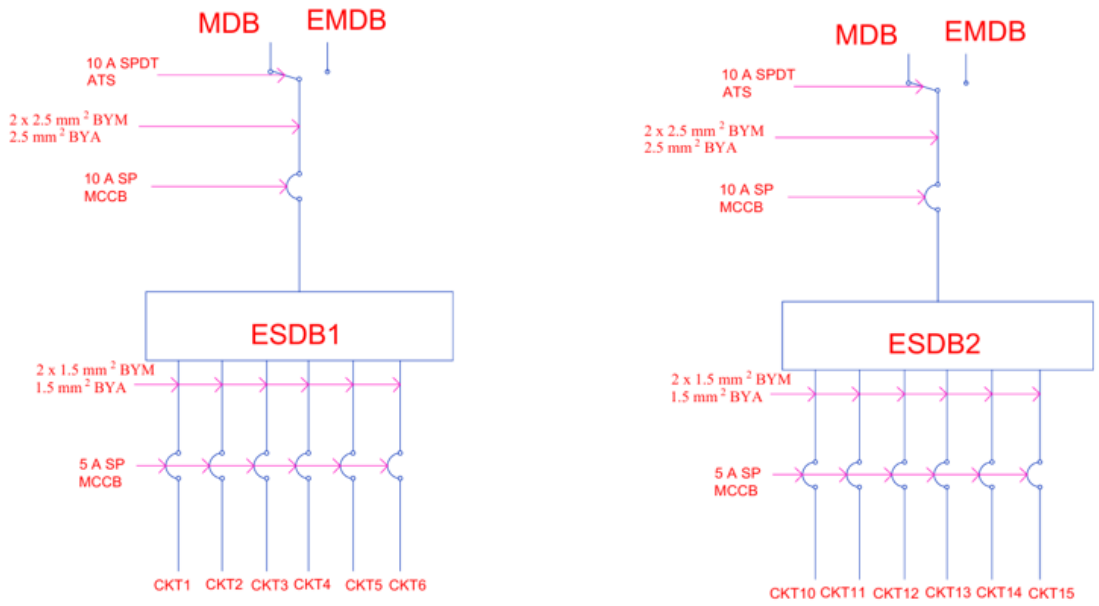


Figure: ESDB Connection diagram (Floor)

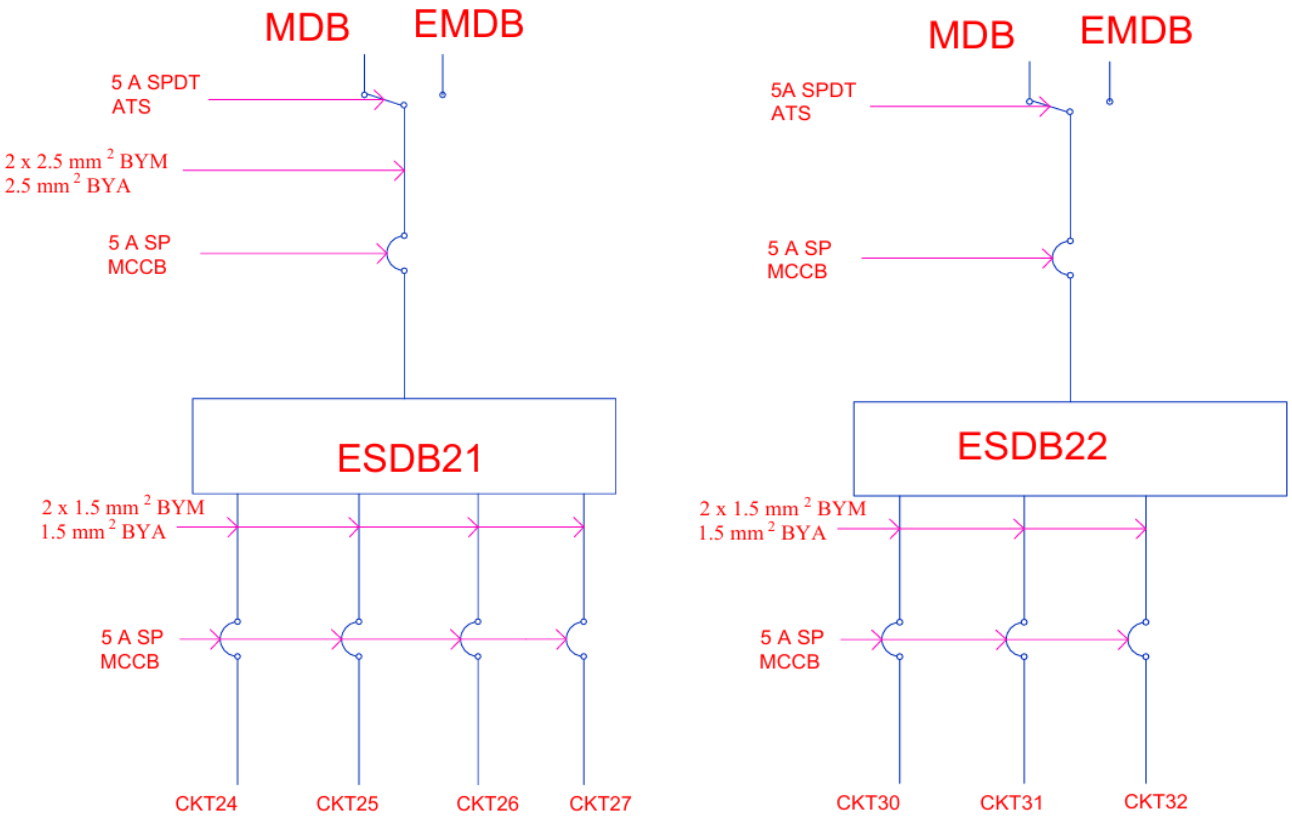


Figure: ESDB Connection Diagram (Ground + Basement)

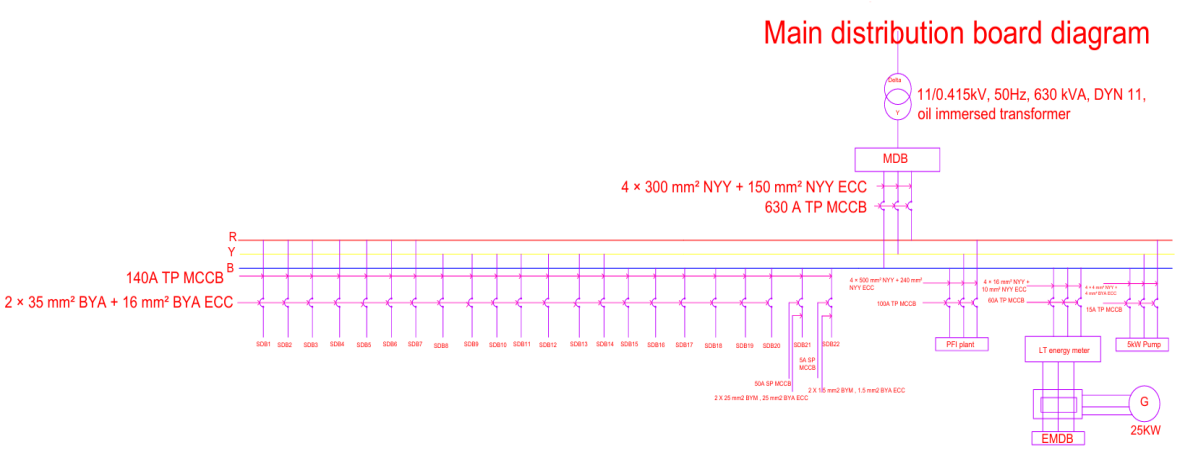


Figure: MDB Connection Diagram

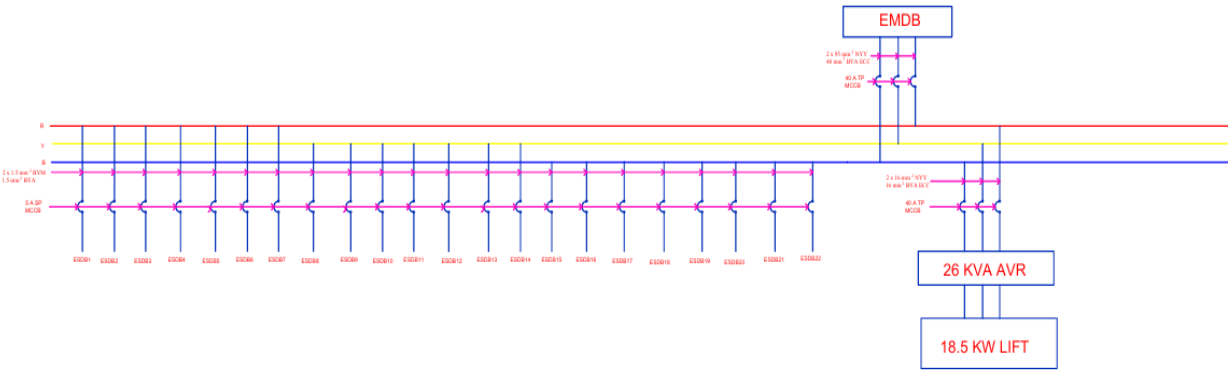


Figure: EMDB Connection Diagram

5.4 Limitations of Tools (PO(e))

The design relies on AutoCAD and manual calculations, limiting real-time analysis and increasing chances of human error. The absence of advanced simulation tools (ETAP, MATLAB) restricts detailed studies such as fault analysis, voltage drop, and protection coordination. This also reduces overall accuracy and limits optimization of the system. Besides, some heavy current running equipment like AC, washing machine, and cooler may require 20A-rated wires, which are omitted here due to simplification.

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

The project follows BNBC 2020 to ensure safety and comfort. Proper illumination supports daily activities and living standards. Backup power and lightning protection maintain essential services during disruptions.

5.5.2 Assessment of Health and Safety Issues

We ensure proper sizing of conductors and circuit breakers, which reduces risks of overload and faults. ESDB ensures power to critical systems during outages. Lightning protection and BNBC compliance enhance overall safety.

5.5.3 Assessment of Legal Issues

Our design complies with BNBC 2020 and the required safety standards. Proper calculations and protection systems help avoid hazards, legal liabilities, and permit issues.

5.6 Sustainability Evaluation (PO(g))

Ours is an energy-efficient design through proper load estimation, diversity factors, and optimized equipment sizing. Minimizes losses, reduces resource use, and ensures long-term reliability.

5.7 Ethical Issues (PO(h))

The engineers' moral duty to put human safety, environmental sustainability, and professional integrity ahead of cost-cutting is the main ethical issue in this electrical infrastructure project. Our design emphasizes safety, compliance, and fair distribution of resources, and maintains transparency in design and promotes energy-efficient, responsible engineering practices.

6. Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

In our project, the group consists of 6 members. People of the group followed the course contents and solved the problems accordingly.

Individual contributions of each members is given below .

Team Member	**Contribution**
2006099	Main distribution board calculation, Basement, Ground, Rooftop layout
2006100	Main floor, Ground, Basement layout, fittings, conduit calculation, design, SLD design.
2006103	Main floor, Ground, Basement layout, fittings, conduit calculation, design, SLD design.
2006104	Rooftop design and protection

2006115	Basement and ground fittings, conduit
2006124	Rooftop design and protection

6.2 Log Book of Project Implementation

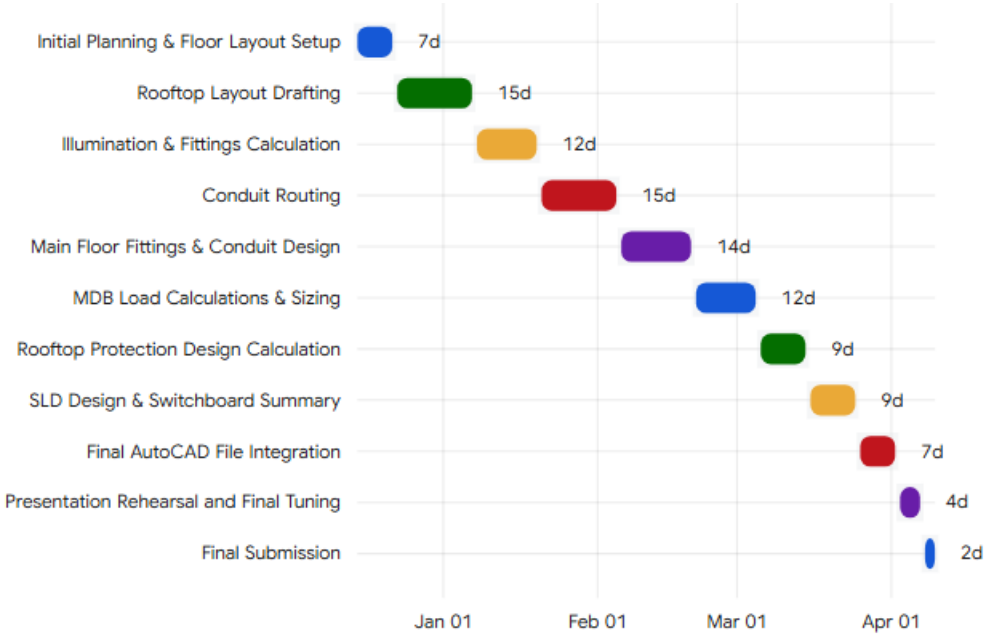
Date	Milestone Achieved	Individual Role (IDs)	Team Role
18/12/2025 (Thu)	Initial Planning & Floor Layout Setup	2006099, 2006100, 2006103	Team drafted the initial architectural base layouts for the Basement, Ground, and Main typical floors in AutoCAD.
09/01/2026 (Fri)	Rooftop Layout Drafting	2006099, 2006100, 2006103	Team completed the rooftop dimensioning and layout to prepare for the protection system setup.
16/01/2026 (Thu)	Illumination & Fittings Calculation (Lower Levels)	2006100, 2006103, 2006115	The team calculated the required lux values and placed lights, fans, and power sockets for the Basement and Ground floor.
23/01/2026 (Thu)	Conduit Routing (Lower Levels)	2006100, 2006103, 2006115	Team routed the main and emergency conduits and calculated wire sizing for Basement and Ground floor.
31/01/2026 (Fri)	Main Floor Fittings & Conduit Design	2006100, 2006103	Team executed fixture placement and full conduit routing for the Main (typical) floors.
19/02/2026 (Thu)	MDB Load Calculations & Sizing	2006099	Member verified load estimations, diversity factors, transformer sizing, and PFI plant requirements.

Date	Milestone Achieved	Individual Role (IDs)	Team Role
26/02/2026 (Thu)	Rooftop Protection Design Calculation	2006104, 2006124	Team performed rolling sphere method calculations to determine the number of air terminals and down conductors.
05/03/2026 (Thu)	SLD Design & Switchboard Summary	2006100, 2006103	Team drafted the Single Line Diagrams (SLD) and connection diagrams for SDBs and ESDBs across all floors.
27/03/2026 (Fri)	Final AutoCAD File Integration	2006099, 2006100, 2006103	Team integrated all layouts, conduit paths, and SLDs into the final master AutoCAD file and checked for overlapping errors.
02/04/2026 (Thu)	Presentation Rehearsal and Final Tuning	All Members	Practiced presentation demo and verified all calculations against BNBC 2020 standards.
03/04/2026 (Fri)	Final Submission	All Members	Submitted the final AutoCAD design files, presentation slides, and evaluation reports.

7. Project Management and Cost Analysis (PO(k))

Timeline of Project Implementation

Based on the times and workflow , we have added a Gantt chart that is shown below –



8. Future Work (PO(l))

As a residential project, it can evolve with technology. Future improvements may include smart monitoring (sensors, smart metering), automated control using BMS, scalability with solar PV integration, and user-friendly apps for energy monitoring and smart home control.

9. References

- Lectures by Yeasir Arafat, Associate Professor, Dept. of EEE, based on BNBC 2020 (Vol. 3, Part VIII, Ch. 1): Electrical and Electronic Engineering Services for Buildings
- AUTOCAD
- https://youtu.be/vUNCu5c4WDE?si=_eQzE5gdRAntzN5O
- https://enel-ukraine.com.ua/en/articles/rolling-sphere-method-for-calculating-lightning-protection-systems-7035